US Semiconductors

State of the Union: raising estimates as AI extends visibility into 2028

Price Objective Change

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Equity
United States
Semiconductors

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Inflation plus stronger demand = higher industry forecasts

We reiterate our thesis of the AI industry moving to addressing structural and physical (chips, power) constraints, from having to defend return-on-investment before. Memory chip shortages and price inflation remain the critical moving pieces, and we update our semis industry models and price objectives to conform to the updated industry estimates. Specifically, we raise our CY30E total semis industry TAM to $2.7Tn, or +28% CAGR between CY25-30E, from $2.3Tn/+23% CAGR prior, led mostly by growth in memory/data center, and also incrementally by recovery in auto/industrial. Our new industry forecasts/ests derive higher price objectives for key semiconductor and semicap equipment companies as summarized in Exhibit 1.

Top 5 themes driving next $1Tn in incremental semi sales

The chip industry took ~50 years to generate its first $1Tn in sales. We expect AI to help add another $1Tn in just the next five years. Key drivers include: **1) AI data center systems** TAM of ~$1.7Tn by CY30E, growing from just ~$273bn in CY25, led by Insatiable demand for compute (see our AI 2030 report); **2) Memory** strength/durability, led by LTAs providing greater confidence in 2-3-year supply/demand/pricing visibility, as seen in recent MU/Anthropic partnership; **3) Semicap/Reshoring/EDA** – clear beneficiaries from extending supply agreements, rising complexity of chips/packaging – see WFE $250bn by CY28E (more below); **4) Analogs** benefiting from rising power reqs (see our AI power semis primer); **5) Agentic CPU** demand, totaling ~$170bn in server opp'ty with benefits across x86/ARM (see our AI CPU TAM report); and **6) Physical AI** over time with also some benefits to DCs for handling more complex queries.

WFE: raising TAM, now see ~$250-300bn in CY28-30E

We tweak our CY27E WFE higher by +4% to $190bn (+31% YoY) from $183bn (+27% YoY) prior while materially raising CY28E by 23% to $250bn (+32% YoY) from $203bn (+11% YoY) prior. Our new CY29/30E WFE forecasts are $268bn (+7% YoY)/$292bn (+9%). The upward revisions are based on our expectation for greater cleanroom availability by CY28, LT visibility afford by memory LTAs, and critical tech inflections which tend to drive WFE-per-wafer higher during upcycles across memory and logic (see analysis in the back). Customer and capacity progress at INTC and Samsung are also positives for advanced F/L while Terafab potential could also potentially be credited.

Core semis benefit from AI, consumer remains a headwind

Of the ~$2.7Tn total semis outlook by CY30E, we see core semis (non-memory) outlook of ~$1.1Tn, growing at ~14% CAGR from ~$567bn in CY25. We see growth driven by server silicon (+24% CAGR) and wired comms (+15%) driven by AI-related chip/networking demand, while consumer-facing PCs (+2%) and smartphones (+0%) remain weak on unit headwinds. We see incrementally better outlook in industrial (+11%) and automotive (+8%) on modest unit recovery and continued content gains.

Key PO changes: MU, semicap/complexity, AI beneficiaries

Based on new industry & WFE forecasts, we raise ests/POs across related beneficiaries.

Exhibit 1: We raise POs across select AI/compute, memory, and semicap companies
PO changes

	PO Changes		
	OLD	NEW	Rating
ACLS	$130	$156	U/P
AEIS	$430	$450	BUY
ALAB	$240	$450	NEUTRAL
AMAT	$540	$720	BUY
ARM	$335	$460	NEUTRAL
CRDO	$252	$340	BUY
INTC	$135	$160	BUY
KLAC	$210	$317	BUY
LRCX	$330	$480	BUY
MKSI	$380	$500	BUY
MRVL	$240	$365	BUY
MU	$950	$1,500	BUY
TER	$365	$525	BUY

U/P = Underperform
Source: BofA Global Research
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Glossary at end of report

Timestamp: 23 June 2026 05:00AM EDT

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Summary of PO changes

Exhibit 2: We raise POs for ACLS, AEIS, ALAB, AMAT, ARM, CRDO, INTC, KLAC, LRCX, MKSI, MRVL, MU and TER. We move to a CY28E valuation basis for multiple companies (from CY27E prior).
Summary of PO changes

	PO Changes			POBR Changes	
	OLD	NEW	Rating	Old POBR	New POBR
ACLS	$130	$156	U/P	27x CY27E PE	26x CY28E PE
AEIS	$430	$450	BUY	36x CY27E PE	32x CY28E PE
ALAB	$240	$450	NEUTRAL	66x CY27E PE	77x CY28E PE (~2.0x PEG)
AMAT	$540	$720	BUY	32x CY27E PE	36x CY28E PE
ARM	$335	$460	NEUTRAL	SOTP CY30E discounted back 2 years (2.0x PEG for IP + 35x PE for Chip)	SOTP CY30E discounted back 2 years (2.5x PEG for IP + 31x PE for Chip)
CRDO	$252	$340	BUY	33x CY27E PE	34x CY28E PE
INTC	$135	$160	BUY	25x CY30E EPS Power, discounted back 2 years	31x CY30E EPS Power, discounted back 2 years
KLAC	$210	$317	BUY	40x CY27E PE	53x CY28E PE
LRCX	$330	$480	BUY	36x CY27E PE	47x CY28E PE
MKSI	$380	$500	BUY	18x CY27E EV/EBIDA	22x CY28E EV/EBITDA
MRVL	$240	$365	BUY	30x CY28E PE	31x CY30E EPS Power, discounted back 2 years
MU	$950	$1,500	BUY	SOTP CY27E (3.1x P/B for trad memory + 27x PE for HBM)	SOTP CY28E (2.5x P/B for trad memory + 31x PE for HBM)
TER	$365	$525	BUY	41x CY27E PE	41x CY28E PE

Source: BofA Global Research

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Global Semis Forecast Update

We model CY26 semis/core semis (ex-memory) growth of +103%/+27% YoY, led by growth in memory, data center, and modestly improved outlook in industrial and automotive, offset by continued unit headwinds in PCs/smartphones/consumer.

By end market, we now model (1) compute and storage up +48% YoY (continued server strength); (2) wireless comms down -12% YoY (smartphone unit headwind); (3) auto sales up +4% on sluggish units but improving content; (4) Industrial up +18% YoY on improved end demand and inventory dynamics since 2H25; (5) consumer down -7% YoY; and (6) wired comms up +29% YoY on data center related infra buildout.

Exhibit 3: We model semis/core semis sales up +103%/+27% YoY in CY26E

Summary of BofA Semiconductor forecasts by end market

Revenue ($mn)	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	CAGR '15-25	CAGR '19-25	CAGR '25-30
Total Semis	**$451**	**$572**	**$594**	**$528**	**$633**	**$787**	**$1,593**	**$2,033**	**$2,194**	**$2,498**	**$2,734**	**8.9%**	**11.1%**	**28.3%**
YoY%	7.7%	27.0%	3.8%	(11.0%)	19.7%	24.3%	102.6%	27.6%	7.9%	13.8%	9.5%			
Memory	**$128**	**$170**	**$150**	**$94**	**$170**	**$220**	**$874**	**$1,184**	**$1,243**	**$1,469**	**$1,646**	**11.0%**	**11.8%**	**49.6%**
YoY%	13.5%	33.4%	(11.9%)	(37.4%)	81.3%	28.9%	297.8%	35.5%	5.0%	18.2%	12.0%			
Core Semis (ex-memory)	**$323**	**$402**	**$444**	**$435**	**$463**	**$567**	**$720**	**$849**	**$951**	**$1,028**	**$1,089**	**8.2%**	**10.8%**	**13.9%**
YoY%	5.6%	24.4%	10.5%	(2.1%)	6.4%	22.6%	27.0%	18.0%	11.9%	8.2%	5.9%			
Compute and Storage	**$108**	**$128**	**$154**	**$167**	**$205**	**$299**	**$441**	**$547**	**$621**	**$674**	**$715**	**13.7%**	**20.7%**	**19.1%**
YoY%	11.2%	19.1%	20.3%	8.4%	22.5%	46.2%	47.5%	24.0%	13.4%	8.7%	6.1%			
PCs	**$56**	**$68**	**$59**	**$52**	**$57**	**$61**	**$56**	**$58**	**$62**	**$66**	**$68**	**4.3%**	**4.3%**	**2.1%**
YoY%	16.5%	22.3%	(12.4%)	(11.8%)	9.3%	7.2%	(9.3%)	4.5%	5.8%	7.2%	3.1%			
Servers (silicon only)	**$27**	**$32**	**$63**	**$80**	**$119**	**$209**	**$356**	**$459**	**$528**	**$577**	**$615**	**28.2%**	**41.0%**	**24.1%**
YoY%	2.2%	16.9%	97.3%	27.5%	48.7%	75.5%	70.6%	28.8%	15.1%	9.2%	6.6%			
Wireless Communications	**$85**	**$104**	**$111**	**$94**	**$93**	**$91**	**$80**	**$80**	**$86**	**$90**	**$93**	**3.3%**	**2.9%**	**0.4%**
YoY%	11.1%	22.3%	6.8%	(15.8%)	(0.5%)	(2.4%)	(11.6%)	(0.2%)	7.1%	4.6%	3.3%			
Smartphone	**$71**	**$88**	**$91**	**$77**	**$80**	**$78**	**$68**	**$67**	**$72**	**$76**	**$78**	**3.9%**	**3.5%**	**0.0%**
YoY%	12.3%	23.7%	3.2%	(15.4%)	3.6%	(2.0%)	(13.0%)	(1.0%)	7.5%	4.7%	3.2%			
Wireless Infrastructure	**$14**	**$16**	**$20**	**$16**	**$13**	**$13**	**$12**	**$13**	**$13**	**$14**	**$14**	**2.9%**	**(0.7%)**	**2.8%**
YoY%	5.0%	14.7%	27.1%	(17.5%)	(20.0%)	(5.0%)	(3.0%)	4.0%	5.0%	4.0%	4.0%			
Automotive	**$34**	**$46**	**$49**	**$59**	**$56**	**$52**	**$54**	**$60**	**$67**	**$73**	**$78**	**5.7%**	**5.6%**	**8.3%**
YoY%	(9.1%)	34.6%	5.4%	22.3%	(6.4%)	(6.0%)	4.1%	10.6%	11.2%	9.0%	6.7%			
Global Automotive Units (mn)	**74.6**	**77.2**	**82.3**	**90.5**	**92.5**	**94.0**	**91.9**	**92.4**	**94.8**	**96.6**	**97.7**	**0.6%**	**0.9%**	**0.8%**
YoY%	(16.1%)	3.5%	6.7%	9.9%	2.2%	1.6%	(2.2%)	0.5%	2.6%	1.9%	1.1%			
Auto semi content ($/LV) / Inv. Adj.	**$459**	**$597**	**$590**	**$657**	**$602**	**$556**	**$593**	**$652**	**$707**	**$756**	**$798**	**5.1%**	**4.6%**	**7.5%**
YoY%	8.4%	30.0%	(1.1%)	11.3%	(8.5%)	(7.5%)	6.5%	10.0%	8.4%	7.0%	5.5%			
Industrial & Other	**$43**	**$54**	**$62**	**$59**	**$45**	**$50**	**$59**	**$66**	**$73**	**$78**	**$82**	**4.1%**	**2.1%**	**10.6%**
YoY%	(2.6%)	26.5%	15.3%	(4.8%)	(24.3%)	10.8%	18.2%	12.5%	9.8%	7.2%	5.6%			
Automation	**$10**	**$13**	**$15**	**$15**	**$12**	**$13**	**$16**	**$18**	**$21**	**$23**	**$25**	**6.4%**	**4.6%**	**13.6%**
YoY%	(3.0%)	29.5%	17.0%	(1.0%)	(15.0%)	5.6%	20.1%	16.6%	13.2%	10.1%	8.5%			
Power/Energy	**$6**	**$7**	**$9**	**$9**	**$8**	**$8**	**$9**	**$10**	**$11**	**$11**	**$12**	**6.8%**	**5.3%**	**7.8%**
YoY%	(0.5%)	25.4%	22.0%	5.0%	(18.0%)	4.1%	13.5%	10.2%	6.2%	5.3%	4.1%			
Consumer	**$35**	**$48**	**$48**	**$30**	**$31**	**$33**	**$31**	**$31**	**$32**	**$33**	**$35**	**1.1%**	**(0.0%)**	**1.3%**
YoY%	6.5%	37.5%	0.1%	(37.1%)	3.4%	4.7%	(6.9%)	(0.1%)	3.1%	6.3%	4.8%			
TVs	**$10**	**$14**	**$14**	**$11**	**$11**	**$11**	**$11**	**$12**	**$12**	**$12**	**$13**	**0.3%**	**2.8%**	**2.8%**
YoY%	10.5%	37.1%	1.8%	(21.7%)	(4.0%)	2.1%	2.3%	2.8%	2.8%	3.0%	3.0%			
Video console SoCs (Gaming)	**$3**	**$6**	**$7**	**$7**	**$3**	**$3**	**$3**	**$3**	**$3**	**$4**	**$5**	**6.5%**	**14.5%**	**7.6%**
YoY%	102.2%	89.9%	26.2%	(4.3%)	(57.1%)	13.0%	(20.2%)	(0.7%)	11.0%	39.2%	17.9%			
Wired Communications	**$18**	**$21**	**$20**	**$25**	**$33**	**$42**	**$54**	**$65**	**$73**	**$80**	**$85**	**9.8%**	**15.2%**	**15.2%**
YoY%	1.0%	17.0%	(8.3%)	26.9%	32.8%	28.1%	29.3%	19.6%	12.2%	9.2%	6.9%			
Ethernet/Network switch	**$5**	**$5**	**$6**	**$9**	**$13**	**$17**	**$25**	**$31**	**$36**	**$40**	**$43**	**12.3%**	**22.6%**	**20.2%**
YoY%	(3.8%)	11.9%	15.0%	45.0%	40.0%	35.0%	42.0%	28.0%	15.0%	11.0%	8.0%			
Optical Equipment	**$4**	**$5**	**$5**	**$6**	**$9**	**$12**	**$15**	**$19**	**$21**	**$24**	**$26**	**18.4%**	**20.5%**	**16.9%**
YoY%	9.8%	11.8%	5.0%	30.0%	35.0%	35.0%	32.0%	20.0%	15.0%	11.0%	8.0%			

Source: BofA Global Research, Mercury Research, Gartner, Omdia, SIA

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In 2026, we expect semis/core semis sales to grow +103%/+27% YoY. Memory is expected to be even stronger, now up nearly +298% YoY (after +81%/+29% YoY growth in CY24/25).

By device type, we see CY26 memory sales driven by both DRAM (+309% YoY) and NAND (+295% YoY).

Ex-memory, microprocessors (MPUs) strong (+33% YoY) on hyperscaler consumption modestly offset by weak PC units, and logic particularly strong (+39% YoY) on AI accelerator-related demand. We also model industrial-centric markets (MCUs, Analog) seeing recovery following inventory digestion throughout 2024 and 1H25. In CY26, we model other markets (discretes, optos, sensors) to generally return to healthy growths (+6% YoY) as well, particularly driven by data center-related optoelectronic demands.

Exhibit 4: We model memory, logic, microcomponents driving semiconductor growth
Summary of BofA Semis Forecast by device type

Revenue ($bn)	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	CAGR '15-25	CAGR '19-25	CAGR '25-30
Total Semis	**$451**	**$572**	**$594**	**$528**	**$633**	**$787**	**$1,593**	**$2,033**	**$2,194**	**$2,498**	**$2,734**	**8.9%**	**11.1%**	**28.3%**
YoY%	7.7%	27.0%	3.8%	(11.0%)	19.7%	24.3%	102.6%	27.6%	7.9%	13.8%	9.5%			
Memory	**$128**	**$170**	**$150**	**$94**	**$170**	**$220**	**$874**	**$1,184**	**$1,243**	**$1,469**	**$1,646**	**11.0%**	**11.8%**	**49.6%**
YoY%	13.5%	33.4%	(11.9%)	(37.4%)	81.3%	28.9%	297.8%	35.5%	5.0%	18.2%	12.0%			
DRAM	**$65**	**$92**	**$78**	**$47**	**$88**	**$134**	**$547**	**$779**	**$838**	**$994**	**$1,136**		**13.5%**	**53.4%**
YoY%	4.6%	40.9%	(15.7%)	(39.1%)	85.9%	52.2%	308.8%	42.4%	7.6%	18.7%	14.2%			
NAND	**$59**	**$73**	**$67**	**$42**	**$78**	**$81**	**$320**	**$397**	**$397**	**$466**	**$501**		**9.8%**	**43.9%**
YoY%	26.4%	25.3%	(8.2%)	(36.9%)	84.0%	3.7%	294.5%	24.1%	0.1%	17.4%	7.3%			
Core Semis (ex-memory)	**$323**	**$402**	**$444**	**$435**	**$463**	**$567**	**$720**	**$849**	**$951**	**$1,028**	**$1,089**	**8.2%**	**10.8%**	**13.9%**
YoY%	5.6%	24.4%	10.5%	(2.1%)	6.4%	22.6%	27.0%	18.0%	11.9%	8.2%	5.9%			
Analog	**$56**	**$74**	**$89**	**$81**	**$80**	**$87**	**$94**	**$102**	**$110**	**$115**	**$119**	**6.7%**	**8.2%**	**6.5%**
YoY%	3.2%	33.1%	20.1%	(8.8%)	(2.0%)	8.8%	8.1%	9.4%	7.1%	4.7%	3.3%			
Microcomponents	**$70**	**$81**	**$79**	**$77**	**$79**	**$85**	**$108**	**$137**	**$168**	**$186**	**$194**	**3.3%**	**4.2%**	**17.9%**
YoY%	4.9%	15.5%	(1.5%)	(3.5%)	3.7%	7.0%	27.0%	27.4%	21.9%	11.1%	4.1%			
Microprocessors	**$51**	**$56**	**$51**	**$45**	**$53**	**$61**	**$81**	**$107**	**$133**	**$149**	**$152**	**3.5%**	**4.1%**	**20.2%**
YoY%	6.5%	10.4%	(9.5%)	(11.2%)	18.0%	14.2%	33.4%	31.6%	24.9%	12.0%	2.0%			
Microcontrollers	**$18**	**$23**	**$27**	**$29**	**$24**	**$23**	**$26**	**$29**	**$31**	**$35**	**$39**	**4.2%**	**4.8%**	**11.2%**
YoY%	3.3%	25.0%	20.8%	7.6%	(19.2%)	(2.4%)	12.0%	11.8%	7.6%	11.3%	13.2%			
Logic	**$118**	**$154**	**$176**	**$178**	**$213**	**$301**	**$418**	**$495**	**$547**	**$592**	**$634**	**12.7%**	**18.9%**	**16.1%**
YoY%	11.1%	30.2%	14.2%	1.4%	19.1%	41.5%	38.8%	18.6%	10.4%	8.2%	7.1%			
Others (Discretes, optos, sensors)	**$79**	**$93**	**$100**	**$98**	**$91**	**$95**	**$101**	**$114**	**$126**	**$136**	**$142**	**4.5%**	**3.1%**	**8.5%**
YoY%	0.3%	17.4%	7.1%	(1.2%)	(7.5%)	4.0%	6.4%	13.4%	10.7%	7.2%	4.9%			

Source: BofA Global Research, Mercury Research, Gartner, Omdia, SIA

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Looking at our forecasts versus BofA Global semis bottom-up estimates, we are generally in line with the growth trajectory for CY26E/CY27E ex-NVDA (which now realizes significant revenue from system sales growth and software). Modest outperformance versus industry reflects continued industry trends of industry consolidation and share gains by top vendors.

Exhibit 5: Our Semis industry outlook generally aligns with bottoms-up company estimates
Summary of BofA semis forecasts versus bottoms-up estimates

Semis ex-mem Forecast ($bn)	2025	2026E	2027E	2028E	2029E	2030E
BofA Semi Ex-Memory Forecast	$567	$720	$849	$951	$1,028	$1,089
YoY (%)	*22.6%*	*27.0%*	*18.0%*	*11.9%*	*8.2%*	*5.9%*
BofA Coverage Bottoms-Up Sales	$539	$767	$981	$1,191	$1,434	$1,583
YoY (%)	*26.8%*	*42.3%*	*27.9%*	*21.4%*	*20.4%*	*10.4%*
BofA Coverage Bottoms-Up Ex-NVDA Sales	$323	$407	$515	$632	$717	$809
YoY (%)	*9.6%*	*25.9%*	*26.8%*	*22.7%*	*13.5%*	*12.8%*

Source: BofA Global Research estimates, SIA, company reports

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Wafer Fab Equipment (WFE) Forecast

We retain our CY26 WFE estimate of $144bn (+24% YoY), but raise CY27-30 figures. Our new CY27/28 forecasts are $190bn (+31% YoY)/$250bn (+32% YoY) from $183bn (+27% YoY) and $203bn (+11% YoY). We expect CY29/30 WFE of $268bn (+7% YoY)/$292bn (+9% YoY).

Our estimates imply 20% CAGR for WFE CY25-30E. We expect strength to be biased towards memory (21% CY25-30E CAGR) which could reach $109bn by CY30 from ~$53bn in CY26. We expect gains to be led by DRAM (22% CAGR) which grows towards 30% of overall WFE but NAND (18% CAGR) is also robust. Non-memory (i.e. foundry/logic) should also see strong growth of 19% CAGR.

Exhibit 6: We estimate WFE of $144bn in CY26E, +24% YoY, and $190bn in CY27E, +31% YoY
Wafer Fab Equipment (WFE) outlook

														CAGR		
	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	'25-'30E	'19-'25	'15-'25
Wafer Fab Equipment (WFE) ($bn)	**$55.2**	**$52.2**	**$61.1**	**$87.8**	**$95.2**	**$97.9**	**$105.3**	**$116.9**	**$144.4**	**$189.8**	**$250.1**	**$268.4**	**$291.7**	**20.1%**	**14.4%**	**14.3%**
YoY	15.8%	(5.4%)	17.1%	43.6%	8.4%	2.9%	7.6%	11.0%	23.5%	31.4%	31.8%	7.3%	8.7%			
Memory WFE ($bn)	**$35.9**	**$20.4**	**$23.8**	**$35.1**	**$34.4**	**$27.4**	**$33.7**	**$42.1**	**$53.4**	**$70.1**	**$92.2**	**$94.4**	**$108.5**	20.9%	12.9%	10.4%
YoY	34.4%	(43.3%)	17.1%	47.3%	(1.9%)	(20.4%)	23.0%	24.9%	27.0%	31.1%	31.6%	2.4%	15.0%			
Memory sales ($bn)	$158.0	$112.5	$127.7	$170.3	$150.0	$94.0	$170.3	$219.6	$873.5	$1,183.9	$1,243.2	$1,469.2	$1,645.7			
Memory WFE capital intensity (% of sales)	23%	18%	19%	21%	23%	29%	20%	19%	6%	6%	7%	6%	7%			
Memory capex ($bn)	$53.9	$45.1	$40.5	$52.3	$59.6	$40.7	$59.1	$74.3	$115.1	$141.0	$153.5	$161.9	$172.7			
DRAM WFE ($bn)	**$12.4**	**$9.2**	**$9.9**	**$15.3**	**$14.0**	**$20.0**	**$26.9**	**$30.2**	**$39.6**	**$52.3**	**$67.4**	**$70.8**	**$82.2**	22.1%	21.8%	12.6%
YoY	42.6%	(25.4%)	7.2%	54.6%	(8.6%)	42.9%	34.7%	12.2%	31.0%	32.0%	29.0%	5.0%	16.0%			
DRAM sales ($bn)	$72.8	$62.5	$65.4	$92.1	$77.7	$47.3	$87.9	$133.8	$547.0	$779.1	$838.0	$994.3	$1,135.9			
DRAM WFE capital intensity (% of sales)	20%	15%	15%	17%	18%	42%	31%	23%	7%	7%	8%	7%	7%			
DRAM capex ($bn)	$18.4	$18.1	$15.9	$23.4	$26.9	$22.8	$37.4	$51.8	$83.0	$91.2	$84.9	$92.4	$101.7			
NAND WFE ($bn)	**$23.0**	**$10.8**	**$13.7**	**$19.5**	**$19.3**	**$6.7**	**$6.3**	**$11.2**	**$13.3**	**$17.3**	**$24.2**	**$23.0**	**$25.8**	18.2%	0.5%	6.2%
YoY	30.5%	(52.8%)	26.1%	42.5%	(0.8%)	(65.5%)	(5.2%)	77.0%	19.0%	30.0%	40.0%	(5.0%)	12.0%			
NAND sales ($bn)	$47.2	$46.3	$58.5	$73.3	$67.3	$42.4	$78.1	$81.0	$319.5	$396.6	$397.1	$466.3	$500.6			
NAND WFE capital intensity (% of sales)	50%	23%	23%	27%	29%	16%	8%	14%	4%	4%	6%	5%	5%			
NAND capex ($bn)	$32.2	$24.1	$21.5	$28.5	$32.2	$17.6	$21.1	$21.9	$31.2	$49.0	$67.7	$68.6	$70.1			
Other Memory WFE ($bn)	**$0.5**	**$0.3**	**$0.3**	**$0.3**	**$1.1**	**$0.7**	**$0.4**	**$0.7**	**$0.5**	**$0.5**	**$0.5**	**$0.6**	**$0.6**	(2.2%)	15.5%	10.4%
YoY	26.9%	(46.4%)	(6.9%)	25.2%	248.5%	(34.7%)	(40.8%)	51.5%	(20.0%)	(10.0%)	7.5%	7.5%	7.5%			
Other Memory sales ($bn)	$49.2	$3.7	$3.8	$4.9	$5.0	$4.2	$4.3	$4.8	$7.0	$8.2	$8.0	$8.6	$9.2			
Other Memory WFE intensity (% of sales)	1%	8%	7%	7%	22%	17%	10%	14%	8%	6%	6%	6%	6%			
Other Memory capex ($bn)	$3.2	$2.9	$3.1	$0.5	$0.4	$0.3	$0.6	$0.6	$0.9	$0.8	$0.8	$0.8	$0.9			
Non-memory WFE ($bn)	**$19.3**	**$31.9**	**$37.3**	**$52.7**	**$61.2**	**$70.5**	**$71.6**	**$74.8**	**$91.0**	**$119.7**	**$157.9**	**$174.0**	**$183.2**	19.6%	15.3%	17.4%
YoY	(7.9%)	64.8%	17.1%	41.3%	16.2%	15.1%	1.6%	4.5%	21.6%	31.6%	31.9%	10.2%	5.3%			
Non-memory semis (excl. Foundry) sales ($bn)	$310.8	$305.9	$322.9	$401.8	$443.9	$434.5	$462.5	$567.0	$719.9	$849.5	$950.8	$1,028.4	$1,088.6			
Non-memory WFE capital intensity (% of sales)	6%	10%	12%	13%	14%	16%	15%	13%	13%	14%	17%	17%	17%			
Non-memory capex ($bn)	$49.2	$54.3	$69.8	$100.3	$123.0	$131.6	$108.3	$118.2	$139.1	$153.2	$166.4	$169.7	$194.3			
Foundry WFE ($bn)	**$11.1**	**$19.3**	**$23.8**	**$36.9**	**$42.6**	**$50.5**	**$51.0**	**$60.5**	**$75.0**	**$101.2**	**$137.1**	**$150.9**	**$158.4**	21.2%	21.0%	19.6%
YoY	(19.0%)	73.3%	23.7%	54.7%	15.6%	18.6%	1.0%	18.5%	24.0%	35.0%	35.5%	10.0%	5.0%			
Foundry sales ($bn)	$63.4	$66.5	$76.3	$100.7	$130.5	$114.7	$137.6	$171.3	$195.1	$216.8	$242.6	$271.4	$0.0			
Foundry WFE capital intensity (% of sales)	18%	29%	31%	37%	33%	44%	37%	35%	38%	47%	57%	56%	#######			
Foundry capex ($bn)	$21.0	$24.2	$34.2	$51.0	$64.9	$58.8	$64.1	$73.3	$95.9	$105.9	$116.2	$115.9	$137.8			
Logic/Intel WFE ($bn)	**$5.3**	**$9.1**	**$9.6**	**$10.3**	**$12.2**	**$12.0**	**$13.3**	**$8.8**	**$11.0**	**$12.7**	**$14.6**	**$16.1**	**$17.2**	14.4%	(0.6%)	12.3%
YoY	19.0%	71.8%	6.2%	6.4%	18.5%	(1.5%)	11.1%	(34.1%)	25.0%	16.0%	15.0%	10.0%	7.0%			
Logic/Intel sales ($bn)	$137.0	$139.3	$153.1	$165.5	$157.4	$163.6	$207.5	$236.9	$259.7	$434.4	$0.5	$0.2	$0.0			
Logic/Intel WFE capital intensity (% of sales)	4%	7%	6%	6%	8%	7%	6%	4%	4%	3%	2,959%	10,294%	#DIV/0!			
Logic/Intel capex ($bn)	$11.7	$15.7	$13.8	$18.7	$24.8	$25.8	$25.1	$17.7	$18.2	$19.6	$20.4	$22.1	$24.0			
IDM/Other WFE ($bn)	**$2.9**	**$3.5**	**$3.8**	**$5.6**	**$6.5**	**$8.0**	**$7.3**	**$5.6**	**$5.0**	**$5.8**	**$6.1**	**$7.0**	**$7.5**	6.2%	8.0%	10.2%
YoY	3.4%	20.2%	8.7%	45.3%	16.4%	23.2%	(8.7%)	(23.4%)	(10.0%)	15.0%	6.0%	15.0%	7.0%			
IDM/Other sales ($bn)	$173.8	$166.6	$169.8	$236.3	$286.5	$270.9	$255.0	$330.0	$460.2	$415.05	$950.27	$1,028.25	$1,088.61			
IDM/Other WFE capital intensity (% of sales)	2%	2%	2%	2%	2%	3%	3%	2%	1%	1%	1%	1%	1%			
IDM/Other capex ($bn)	$16.5	$14.4	$21.8	$30.7	$33.2	$47.0	$19.1	$27.2	$25.1	$27.7	$29.8	$31.7	$32.5			

Source: BofA Global Research estimates, SIA, IMF, Gartner

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Leading-edge, non-China, and EUV driving future growth

We expect leading-edge WFE (sub-22nm) to grow 22% CAGR CY25-30E with large growth years in CY26-28 as TSMC seems likely to raise capex and sub-5nm investments broaden across the foundry/IDM ecosystem (Samsung, INTC, Rapidus) to support AI/HPC demand. Relatedly, we expect much of this growth to benefit EUV growing at 22% CAGR. In contrast to CY22-25E, we expect a geo mix shift from China to non-China as reshoring projects around the world ramp up, fueling a 24% CY25-30E CAGR, while China WFE grows 10% CAGR as the market digests existing capacity.

Exhibit 7: Breakdown of WFE estimates by trailing-edge vs. leading-edge, non-litho, China vs. non-China spending

We expect CY25E to likely be a strong year for leading-edge and non-China WFE

												CAGRs		
WFE ($bn)	**2020**	**2021**	**2022**	**2023**	**2024**	**2025**	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**	**'25-'30E**	**'20-'25E**	**'15-'25**
Total WFE	**61.1**	**87.8**	**95.7**	**97.9**	**105.3**	**116.9**	**144.4**	**189.8**	**250.1**	**268.4**	**291.7**	**20.1%**	**13.8%**	**14.3%**
YoY	**17.1%**	**43.6%**	**9.0%**	**2.3%**	**7.6%**	**11.0%**	**23.5%**	**31.4%**	**31.8%**	**7.3%**	**8.7%**			
Memory	**23.8**	**35.1**	**34.4**	**27.4**	**33.7**	**42.1**	**53.4**	**70.1**	**92.2**	**94.4**	**108.5**	**20.9%**	**12.0%**	**10.4%**
Foundry/Logic/IDM	**37.3**	**52.7**	**61.2**	**70.5**	**71.6**	**74.8**	**91.0**	**119.7**	**157.9**	**174.0**	**183.2**	**19.6%**	**14.9%**	**17.4%**
YoY	17.1%	41.3%	16.2%	15.1%	1.6%	4.5%	21.6%	31.6%	31.9%	10.2%	5.3%			
% of total WFE	61.0%	60.0%	64.0%	72.0%	68.0%	64.0%	63.0%	63.1%	63.1%	64.8%	62.8%			
Leading-edge (<=22nm)	**17.3**	**26.3**	**30.9**	**38.3**	**31.4**	**39.3**	**53.1**	**76.4**	**98.6**	**101.5**	**107.6**	**22.3%**	**17.8%**	**18.9%**
YoY	20.6%	52.2%	17.4%	24.0%	(18.0%)	25.0%	35.0%	44.0%	29.0%	3.0%	6.0%			
% of total WFE	28.3%	30.0%	32.3%	39.2%	29.9%	33.6%	36.7%	40.3%	39.4%	37.8%	36.9%			
Trailing-edge (>22nm)	**20.0**	**26.3**	**30.3**	**32.1**	**40.2**	**35.5**	**37.9**	**43.3**	**59.3**	**72.5**	**75.5**	**16.3%**	**12.2%**	**16.0%**
YoY	14.2%	31.8%	15.1%	6.0%	25.0%	(11.6%)	6.7%	14.3%	37.0%	22.1%	4.2%			
% of total WFE	32.7%	30.0%	31.7%	32.8%	38.1%	30.4%	26.2%	22.8%	23.7%	27.0%	25.9%			
EUV	**5.1**	**7.4**	**7.5**	**10.1**	**9.0**	**13.6**	**18.2**	**26.4**	**29.8**	**34.1**	**36.6**	**21.9%**	**21.5%**	**68.1%**
YoY	64.7%	44.5%	0.6%	35.5%	-10.4%	50.3%	33.9%	45.1%	12.7%	14.5%	7.5%			
% of total WFE	8.4%	8.4%	7.8%	10.3%	8.6%	11.6%	12.6%	13.9%	11.9%	12.7%	12.6%			
Total ex-EUV	**56.0**	**80.4**	**88.2**	**87.8**	**96.3**	**103.3**	**126.2**	**163.4**	**220.3**	**234.3**	**255.0**	**19.8%**	**13.0%**	**13.0%**
YoY	14.0%	43.6%	9.7%	-0.5%	9.7%	7.3%	22.2%	29.4%	34.9%	6.3%	8.8%			
% of total WFE	91.6%	91.6%	92.2%	89.7%	91.4%	88.4%	87.4%	86.1%	88.1%	87.3%	87.4%			
China	**15.5**	**21.4**	**21.6**	**31.7**	**38.1**	**39.2**	**42.6**	**47.6**	**51.3**	**59.8**	**62.7**	**9.9%**	**20.4%**	**26.6%**
YoY	34.1%	38.6%	0.6%	47.2%	20.0%	2.8%	8.8%	11.8%	7.6%	16.7%	4.8%			
% of total WFE	25.3%	24.4%	22.5%	32.4%	36.2%	33.5%	29.5%	25.1%	20.5%	22.3%	21.5%			
Total Non-China	**45.7**	**66.4**	**74.1**	**66.1**	**67.2**	**77.7**	**101.8**	**142.1**	**198.8**	**208.5**	**229.0**	**24.1%**	**11.2%**	**11.2%**
YoY	12.2%	45.3%	11.7%	(10.8%)	1.6%	15.6%	31.0%	39.6%	39.9%	4.9%	9.8%			
% of total WFE	74.7%	75.6%	77.5%	67.6%	63.8%	66.5%	70.5%	74.9%	79.5%	77.7%	78.5%			

Source: BofA Global Research estimates, company data, Gartner, SIA, IMF

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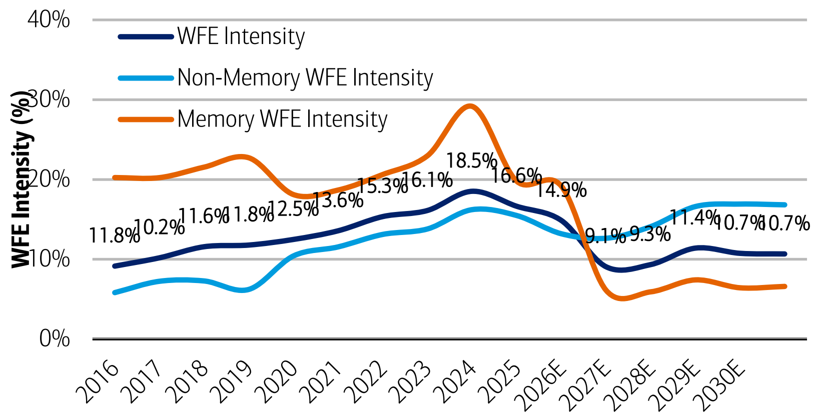
The case for $250bn WFE by CY28E

We present case for WFE spending to reach $250bn by CY28E below. In our view, the combination of unit-driven industry sales gain by the end of the decade, tech inflections, and new capacity unlock should drive WFE to historically-high levels.

WFE intensity: ostensibly low and a less useful metric for this upcycle

A point of contention for investors is the collapse in WFE intensity anticipated from CY27 onwards from the typical mid-teens % range towards historic lows of 9-12%. We see this is mainly led by memory where dramatic ASP gains are artificially depressing intensity, making it a less useful metric to understand how WFE should trend during this upcycle. While we only model overall WFE intensity of ~11% by CY30E, we expect it to eventually recover to mid-teens % levels as units begin to drive industry growth.

Exhibit 8: Overall WFE intensity could decline to historical lows given ASP-driven sales gains
WFE intensity CY16-2030



Source: BofA Global Research estimates, Gartner, company reports

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Analyzing six different semi industry upcycles since CY00 of varying duration, we see that WFE gained in all but one of them and outgrow total semi sales through-cycle in all but two. We note WFE intensity tends to increase a median of ~300bps through-cycle as IDMs/foundries typically perform upgrades and tech transitions during periods of strong industry growth in addition to capacity expansion. However, most of these historical semi cycles were unit-driven rather than ASP-driven aside from CY20-22 and CY24-25, the latter of which has seen a -400bps decline in WFE intensity to-date, partially impacted by the normalization of China spending. While elevated pricing may continue to depress WFE intensity over the next several years, we do expect units to eventually recover as a driving force as capacity comes online.

Exhibit 9: While each industry upcycle has differed in duration, unit vs. ASP contribution, and fundamental drivers, we observe that in most cases WFE intensity tends to rise during the period
WFE performance during every semi industry upcycle CY00-25

# Years	Semi Upcycle	Total Semi Sales Gain	Total WFE Gain	Total Unit Gain	Total ASP Gain	Semi Sales CAGR	WFE CAGR	Unit CAGR	ASP CAGR	Avg. WFE Intensity during period	Change in WFE intensity	Change in WFE Intensity per Wafer	WFE per wafer CAGR
5	CY02-07	81.7%	139.3%	72.9%	5.1%	12.7%	19.1%	11.6%	1.0%	**12.6%**	**365bps**	**NA**	**NA**
2	CY10-11	32.3%	180.5%	24.8%	6.0%	15.0%	67.5%	11.7%	3.0%	**9.8%**	**664bps**	**NA**	**NA**
2	CY13-14	15.2%	0.6%	13.9%	1.1%	7.3%	0.3%	6.7%	0.5%	**9.5%**	**-134bps**	**-12.9%**	**-6.7%**
3	CY16-18	39.9%	80.2%	27.6%	9.6%	11.8%	21.7%	8.5%	3.1%	**10.7%**	**264bps**	**38.0%**	**11.3%**
3	CY20-22	39.1%	83.2%	17.6%	18.3%	11.6%	22.4%	5.6%	5.8%	**14.4%**	**363bps**	**56.6%**	**2.3%**
2	CY24-25	44.7%	19.4%	44.7%	43.5%	20.3%	9.3%	20.3%	19.8%	**16.7%**	**-366bps**	**8.2%**	**2.3%**
										Median	313bps	23.1%	2.3%

Source: BofA Global Research estimates, Gartner, company reports

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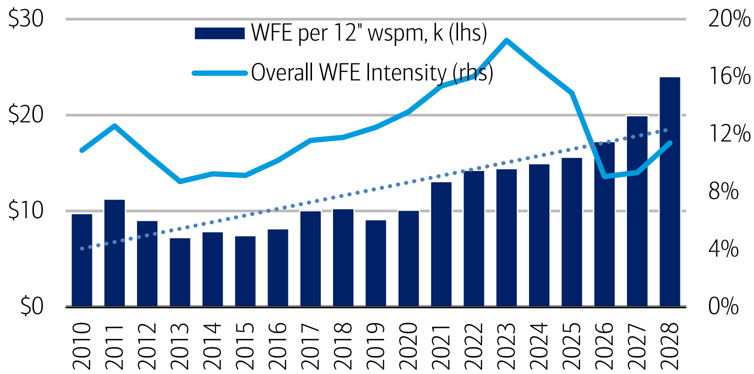
WFE-per-wafer: structurally increasing through-cycle

We think the more relevant metric to track is WFE per 12” wafer start which strips out pricing dynamics and focuses on the investment needed to produce the next unit of output. We see that WFE-per-wafer has steadily increased through various cycles as manufacturing complexity rises and based on historical through-cycle gains, we anticipate the trend will remain intact for overall WFE CY26-28 as massive capacity expansions and unique tech inflections drive higher levels of investment despite a drastic step-down in WFE intensity as a % of industry sales.

Exhibit 10: WFE per 12” wafer has been structurally rising through various cycles and we expect the same CY25-28 despite a sharp decline in WFE intensity

Overall WFE per 12” wafer (lhs) and WFE intensity % (rhs)



Source: BofA Global Research estimates, Gartner, company reports

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Logic WFE/wafer: rises over time but locally peaks 1-2 years into volume ramp

We look at five different upcycles for core semis (ex-memory), i.e foundry/logic semis. Almost all cycles are unit-led but intensity trends have never consistently increased through-cycle as it is highly dependent on the relative maturity of each subsequent node. Each new node tends to reset per-wafer cost upward at launch but WFE per wafer tends to peak 1-2 years into the ramp when yields are low, tools are least productive, and capacity is largely greenfield before normalizing as the node matures. During tech transitions like EUV or the ramp of 28nm high-K metal gate, WFE per wafer grew above +20% CAGR.

Exhibit 11: WFE intensity and WFE per wafer trends tend to be less consistent across foundry/logic cycles but typically see strong expansion during tech transitions such as the move to EUV in CY18-22

History of foundry/logic semi sales (ex-memory) cycles since CY09 and corresponding WFE trends

# Years	Core Semi Upcycle	Core Semi Sales GAR	WFE CAGR	Unit CAGR	ASP CAGR	Avg. WFE Intensity during period	Change in WFE intensity	Change in WFE intensity per wafer	WFE per wafer CAGR	Notes
2	CY09-11	14.7%	-46.1%	11.7%	2.6%	20.2%	(3,444)	51.6%	23.1%	Global Financial Crisis recovery, minimal greenfield, 28nm high-K metal gate ramp
2	CY12-14	4.6%	-13.5%	6.9%	-2.1%	8.0%	(309)	-45.3%	-26.1%	28nm capacity maturing, short-lived 20nm ramp, efficient brownfield adds, booming smartphone-led sales
3	CY15-18	6.4%	8.8%	8.7%	-2.1%	6.6%	40	13.8%	4.4%	FinFET transition, 16/14nm ramp in CY15 and peaking CY16/17
4	CY18-22	9.3%	33.4%	2.3%	6.9%	11.0%	758	140.1%	24.5%	EUV adoption, introduction of 7nm and rapid pacing to 5nm/3nm
2	CY23-25	29.3%	3.0%	20.2%	7.5%	15.0%	(302)	-11.2%	-5.8%	AI accelerator demand ramping on scaled nodes (N5/N4/N3), maturing productivity, limited 2nm contribution
						Median	-302bps	13.8%	4.4%	

Source: BofA Global Research estimates, Gartner, company reports

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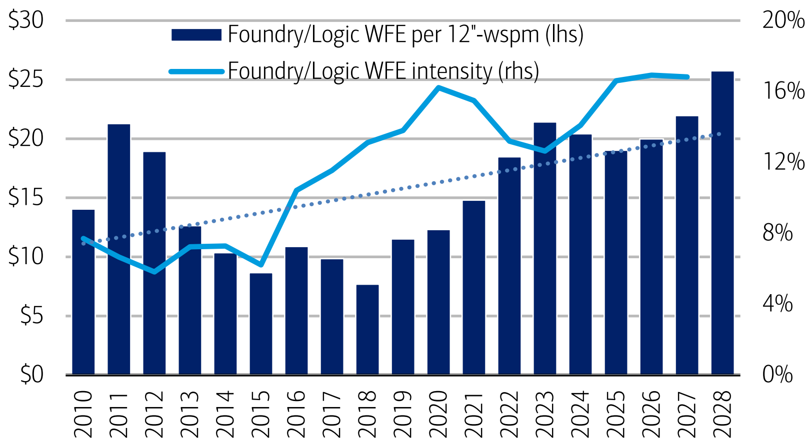
Ahead, we expect strong expansion CY26-28 as 2nm gate-all-around ramps, especially at emerging foundries, where yields are likely low and tools less productive. We also expect a wave of greenfield capacity expansion. High-NA EUV adoption could also be a critical driver in addition to backside power adoption.

Exhibit 12: Through-cycle WFE per wafer gains have been inconsistent for foundry/logic but generally have structurally increased over time via node migrations

Foundry/Logic WFE per 12" wafer start per month (lhs) and foundry/logic WFE intensity (rhs)



Source: BofA Global Research estimates, Gartner, company reports

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DRAM WFE/wafer: litho insertion sharply increasing intensity

Absolute DRAM WFE intensity and WFE per wafer has increased virtually every DRAM sales upcycle since CY09 with the exception of CY23-25 where intensity collapsed following a sharp digestion in China investments. On average, WFE per wafer has increased 22% CAGR each cycle and WFE intensity adds ~300bps. The most important inflections for intensity was when multipatterning progress began to slow and EUV insertion became necessary. The rise of HBM with its high trade ratio (3-4 wafers for every 1 wafer of traditional DRAM), related advanced packaging, higher layer count stacking, and larger capacitors are also drivers.

Exhibit 13: DRAM intensity and WFE per wafer generally increases during DRAM sales cycles even if growth is mainly ASP-led with the exception being CY23-25

History of DRAM sales cycles since CY09 and corresponding WFE trends

# Years	DRAM Upcycle	DRAM Sales GAR	WFE CAGR	Unit CAGR	ASP CAGR	Avg. WFE Intensity during period	Change in WFE intensity	Change in WFE intensity per wafer	WFE per wafer CAGR	Notes
1	CY09-10	74.9%	74.9%	19.8%	46.0%	25.9%	790	27.4%	27.4%	Global Financial Crisis Recovery
2	CY12-14	34.0%	48.6%	-1.8%	36.5%	12.3%	267	25.0%	11.8%	Recovery post Elpida bankruptcy, sub-20nm planar multipatterning
2	CY16-18	55.3%	48.3%	-1.3%	57.3%	15.1%	615	108.2%	44.3%	multipatterning begins to hit a wall, capacitors, greenfield builds
2	CY19-21	22.0%	28.7%	12.6%	8.4%	15.5%	184	48.1%	21.7%	EUV debut, continued shrinks, buried-wordline, deepercpacitor
2	CY23-25	118.4%	23.0%	16.6%	-22.0%	31.8%	(1,969)	31.8%	14.8%	HBM, TSV, die-stacking, thermocompression bonding
						Median	267bps	31.8%	21.7%	

Source: BofA Global Research estimates, Gartner, company reports

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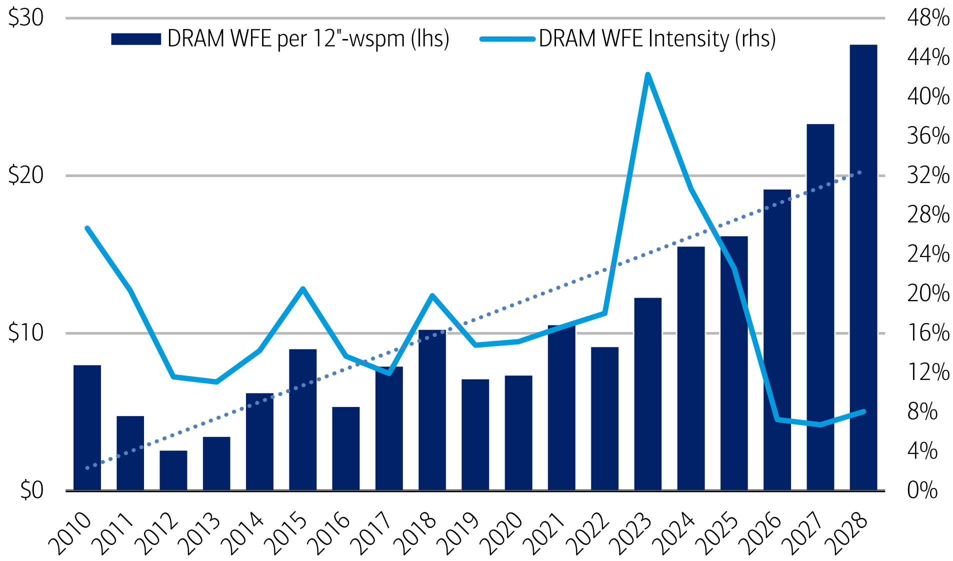
Looking ahead, we anticipate DRAM WFE per wafer to steadily rise as HBM becomes a larger part of the mix with the trade ratio with conventional DRAM becoming even more stark as we move to higher-layer HBM4/5. Larger core dies and the shift from thermocompression bonding to hybrid bonding should also step up equipment adoption while capacitor aspect-ratios continue to climb and EUV insertion deepens. 4F squared DRAM could also drive intensity higher CY28 via greater etch/deposition.

Exhibit 14: Despite a collapse in WFE intensity to historical lows, we expect WFE per wafer to steadily increase through CY28

DRAM WFE per 12" wspm (lhs) and WFE intensity (rhs)



Source: BofA Global Research estimates, Gartner, company reports

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NAND WFE/wafer: migration to higher layer counts drives intensity

During the 2010s, NAND investments exploded to address the surge in mobile demand but we also saw a sharp rise in WFE intensity as structures went vertical with the advent of 3D NAND which required greater etch/dep intensity. Over the last five years, the industry saw limited greenfield investments but steady brownfield spending as the install base saw continuous conversions to higher layer count devices to satisfy bit demand.

Exhibit 15: NAND WFE intensity per wafer has always increased in the past three cycles as bit growth is driven by migrations and conversions rather than greenfield capacity adds

History of NAND sales cycles since CY09 and corresponding WFE trends

# Years	Memory Upcycle	NAND sales CAGR	WFE CAGR	Unit CAGR	ASP CAGR	Avg. WFE Intensity during period	Change in WFE intensity	Change in WFE Intensity per Wafer	WFE per wafer CAGR	Notes
9	CY09-18	15.5%	22.0%	12.1%	3.0%	28.5%	2,375	107.2%	8.4%	SSD/mobile demand, 3D NAND transition
2	CY19-21	18.0%	34.0%	9.1%	8.1%	24.5%	316	76.4%	32.8%	migrations w/ limited greenfield, bit growth coming from higher layers (96L to 176L), multi-deck stacking,
2	CY23-25	77.5%	29.5%	10.8%	60.1%	466.2%	(190)	84.9%	36.0%	Migration-led cycle, trough of CY22-23 NAND downturn, move to 300L, double/triple-decks
						Median	316	84.9%	32.8%	

Source: BofA Global Research estimates, Gartner, company reports

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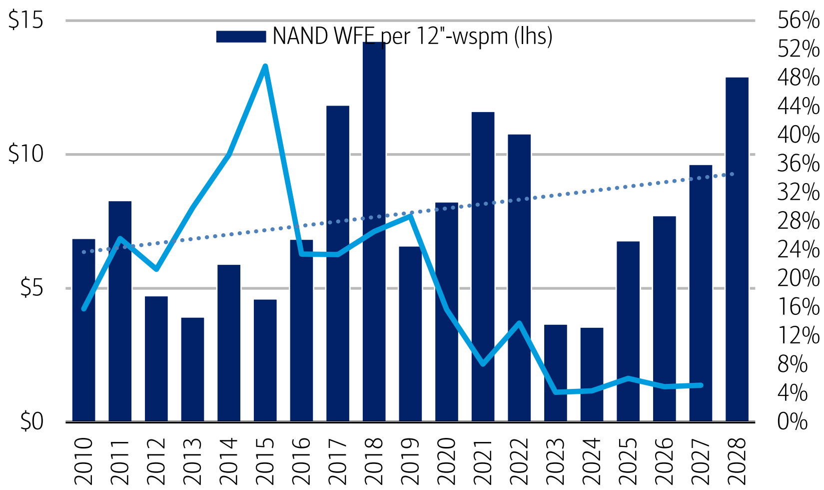
While there could be greenfield activity by 2H27/CY28, we expect most of the WFE per wafer gains to come from continued layer count migrations to 300L/400L, moving to triple deck stacking architecture, and leveraging of deeper high-aspect ratio channel etch. Total expansion could be more moderate versus prior cycles as the market is firmly off its cyclical trough but demand from high-capacity QLC for AI inference storage could be an upside risk.

Exhibit 16: While trends have been choppy, NAND WFE per wafer rises during periods of heightened migration/conversion activity

NAND WFE per 12" wafer start per month (lhs) and NAND WFE intensity (rhs)



Source: BofA Global Research estimates, Gartner, company reports

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Exhibit 17: At the aggregate level, memory has undergone 4 different upcycles over the last 16 years which have always been led by ASP gains. However, we note that WFE intensity increased in each upcycle aside from CY23-25 as tech inflections and units eventually drove growth.

History of memory sales cycles since CY09 and corresponding WFE trends

# Years	Memory Upcycle	Memory Sales CAGR	WFE CAGR	Unit CAGR	ASP CAGR	Avg. WFE Intensity during period	Change in WFE intensity	Change in WFE Intensity per Wafer	WFE per wafer CAGR
2	CY12-14	17.9%	37.6%	4.1%	13.3%	14.7%	355	68.3%	29.7%
2	CY16-18	43.4%	52.0%	-0.2%	43.8%	21.5%	249	102.0%	42.1%
2	CY19-21	20.2%	31.3%	8.7%	10.6%	19.1%	251	63.0%	27.7%
2	CY23-25	99.6%	23.9%	21.4%	64.4%	22.7%	(1,000)	52.2%	23.4%
						Median	250bps	65.7%	28.7%

Source: BofA Global Research estimates, Gartner, company reports

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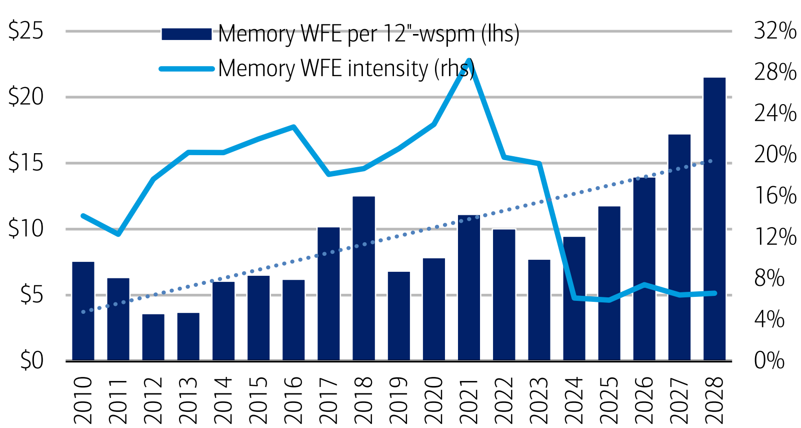
Exhibit 18: Taken together across both DRAM and NAND, overall memory WFE per wafer should structurally increase into CY26-28 despite unprecedented ASP gains leading to lower intensity

Overall memory WFE per 12" wafer start per month (lhs) and WFE intensity (rhs)



Source: BofA Global Research estimates, Gartner, company reports

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WFE per wafer analysis implies $250bn WFE by CY28E

Using our justifications above and increasing WFE per wafer by trends consistent with periods of aggressive node pacing, brownfield and greenfield capacity expansions, and tech transitions in the past, we can come up with implied WFE using our estimates for logic and memory wafers.

We expect the biggest gains in memory WFE per wafer but also expect modest growth in foundry/logic. In total, implied WFE could reach $193bn in CY27 and $245bn by CY28 directionally supporting our updated WFE estimates of $190bn and $250bn.

Exhibit 19: Using our forecasts for logic and memory wafers and assuming WFE per wafer gains in-line consistent with historical precedent, we calculate implied WFE of $193bn in CY27E and $245bn by CY28E supporting our updated forecasts

Translating WFE per wafer trends into aggregate potential WFE through CY28

	2023	2024	2025	2026	2027	2028	CY26-28 CAGR
NAND wafers 12" wspm K	1,820	1,781	1,651	1,726	1,726	1,878	4%
NAND WFE per 12" wspm	$4	$4	$7	$8	$10	$13	30%
NAND WFE	**$6,667**	**$6,320**	**$11,184**	**$13,309**	**$17,301**	**$24,472**	**36%**
YoY Growth		*-5.2%*	*77.0%*	*19.0%*	*30.0%*	*41.5%*	
DRAM wafers 12" wspm K	1,628	1,734	1,867	2,066	2,242	2,379	7%
DRAM WFE per 12" wspm	$12	$16	$16	$19	$23	$28	22%
DRAM WFE	**$20,002**	**$26,943**	**$30,237**	**$39,611**	**$52,314**	**$67,536**	**31%**
YoY Growth		*34.7%*	*12.2%*	*31.0%*	*32.1%*	*29.1%*	
Memory wafers 12" wspm K	3,651	3,758	3,892	4,092	4,269	4,407	4%
Memory WFE per 12" wspm	$7	$9	$11	$13	$16	$21	27%
Memory WFE	**$26,670**	**$33,263**	**$41,421**	**$52,919**	**$69,615**	**$92,009**	**32%**
YoY Growth		*24.7%*	*24.5%*	*27.8%*	*31.5%*	*32.2%*	
Leading-edge F/Lwafers 12" wspm K	226	349	467	639	838	983	24%
Leading-edge F/LWFE per 12" wspm	$170	$90	$84	$83	$91	$100	10%
Leading-edge F/L WFE	**$38,347**	**$31,445**	**$39,306**	**$53,063**	**$76,559**	**$98,780**	**36%**
YoY Growth		*-18.0%*	*25.0%*	*35.0%*	*44.3%*	*29.0%*	
Trailing-edge F/L wafers 12" wspm K	3,064	3,158	3,465	3,912	4,612	5,150	15%
Trailing-edge F/L WFE per 12" wspm	$10	$13	$10	$10	$10	$11	5%
Trailing-edge F/L WFE	**$32,128**	**$40,166**	**$35,507**	**$37,899**	**$46,923**	**$55,015**	**20%**
YoY Growth		*25.0%*	*-11.6%*	*6.7%*	*23.8%*	*17.2%*	
Total Foundry/Logic wafers 12" wspm K	3,290	3,507	3,932	4,551	5,450	6,133	16%
Total Foundry/Logic WFE per 12" wspm	$21.4	$20.4	$19.0	$20.0	$22.7	$25.1	12%
Total Foundry/Logic WFE	**$70,475**	**$71,611**	**$74,813**	**$90,962**	**$123,482**	**$153,795**	**30%**
YoY Growth		*1.6%*	*4.5%*	*21.6%*	*35.8%*	*24.5%*	
WFE wafers 12" wspm K	6,941	7,265	7,824	8,643	9,720	10,540	10%
WFE per 12" wspm	$14	$14	$15	$17	$20	$23	18%
WFE	**$97,145**	**$104,873**	**$116,234**	**$143,881**	**$193,097**	**$245,803**	**31%**
YoY Growth		*8.0%*	*10.8%*	*23.8%*	*34.2%*	*27.3%*	

Source: BofA Global Research estimates, Gartner, company reports

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MU: Raise PO to $1,500, Buy

We maintain our bullish views for AI memory, supported by robust demand and limited supplies through CY26-28.

As we pointed out in our 2026 AI inference primer report, memory capacity and bandwidth are increasingly becoming a bottleneck in AI inference, as opposed to historically compute most often being the bottleneck.

This is especially true for long-context, high-concurrency, multi-step inference, as every generated token must repeatedly access model weights and the accumulated KV cache, while the KV cache grows with sequence length, batch size, number of layers, and model dimensionality.

Moreover, agentic systems require persistent state, and while compute (raw FLOPs) scales once per model, memory (bandwidth) scales per user, per query, and per agent.

Memory content per AI system scales faster than compute

Particularly, we flag AI model context windows are expanding much faster than model compression can offset. In other words, the raw demand for memory is increasing faster than various software/hardware abilities to compress and/or distill the amount of memory used during the inference process.

For instance, NVIDIA's software work on NVFP4 KV cache quantization aims to free HBM capacity to enable larger batches, longer sequences, and higher cache-hit rates. Yet, the raw demand for HBM memory is growing much faster than the ability to supply HBM.

Overall, we forecast HBM TAM to grow to ~$246bn by CY30 from ~$35bn in CY25, growing at +34% CAGR. We also expect HBM content per accelerator to grow to ~464GB by CY30 from ~187GB in CY25, growing at ~18% CAGR.

For HBM pricing, we expect it to reach ~$17.5/GB in CY27-28, vs. ~$14.3/GB in CY26.

Exhibit 20: We see HBM content/accelerator to grow to ~464GB/accelerator by CY30 from ~187GB/accelerator in CY25

HBM Content Outlook per Accelerator

	2021	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	CAGR '26-'30
HBM TAM ($bn)	$1	$2	$4	$17	$35	$77	$135	$176	$205	$246	34%
HBM Units (GB bn)	0.05	0.10	0.47	1.58	3.06	5.38	7.74	10.08	12.85	16.19	32%
Accelerator TAM (units mn)	4.8	4.9	6.2	11.0	16.3	22.6	27.4	28.2	32.0	34.9	11%
Total HBM Content / XPU	**10 GB**	**20 GB**	**76 GB**	**143 GB**	**187 GB**	**238 GB**	**282 GB**	**358 GB**	**402 GB**	**464 GB**	18%
HBM ASP ($/GB)	**$21.1**	**$16.7**	**$8.8**	**$11.0**	**$11.3**	**$14.3**	**$17.4**	**$17.5**	**$15.9**	**$15.2**	2%
HBM TAM ($bn)	$1	$2	$4	$17	$35	$77	$135	$176	$205	$246	34%
Accelerator TAM ($bn)	$9	$14	$45	$120	$196	$381	$603	$830	$1,026	$1,171	32%
HBM as % of XPU (%)	**12%**	**11%**	**9%**	**14%**	**18%**	**20%**	**22%**	**21%**	**20%**	**21%**	

Source: BofA Global Research estimates

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Notably, NVDA's latest Vera Rubin (2H26) systems use 288GB HBM4 per accelerator.

Exhibit 21: A Vera Rubin-based system roughly requires 288GB of HBM, 3.4TB of LPDDR, and ~1.8GB of SRAM memory per accelerator

Vera Rubin System Memory (HBM, LPDDR, SRAM) Requirement per Accelerator

	Vera Rubin	Vera Rubin	Vera CPU Rack	Groq-3 LPU	On-Chip Cache
Type	HBM4	LPDDR5X	LPDDR5X	SRAM	SRAM
Memory per chip	**288 GB**	**1,500 GB**	**1,500 GB**	**0.5 GB**	
Chips per rack	72	36	256	256	
Memory per rack	21 TB	54 TB	384 TB	0.128 TB	

Exhibit 21: A Vera Rubin-based system roughly requires 288GB of HBM, 3.4TB of LPDDR, and ~1.8GB of SRAM memory per accelerator
Vera Rubin System Memory (HBM, LPDDR, SRAM) Requirement per Accelerator

	Vera Rubin	Vera Rubin	Vera CPU Rack	Groq-3 LPU	On-Chip Cache
Racks per pod	16	16	8	8	
Memory per pod	**332 TB**	**864 TB**	**3,072 TB**	**1.024 TB**	**~1.024 TB**
Memory per Accelerator	**288 GB**	**750 GB**	**2,667 GB**	**0.889 GB**	**~0.889 GB**

Source: BofA Global Research estimates

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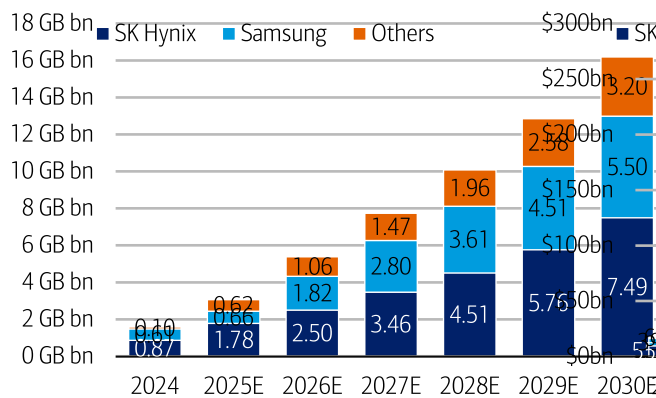
Exhibit 22: Industry HBM shipment could reach 16.19[illegible] CY30E, growing at +40% CAGR between CY25-30E
HBM Unit Shipment Forecast by Vendor (GB bn)



Source: BofA Global Research estimates

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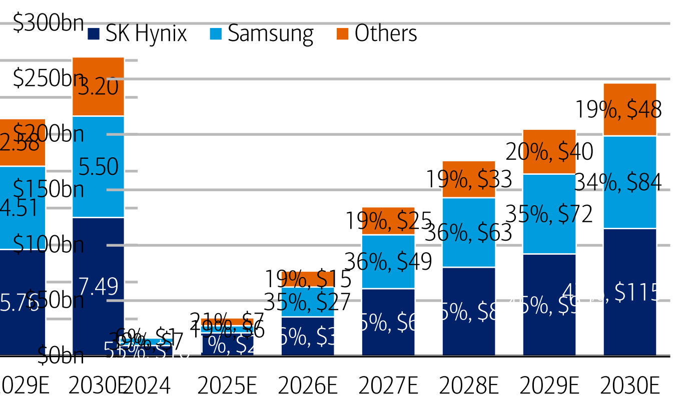
Exhibit 23: Industry HBM sales could reach $168bn by CY30E, growing at +37% CAGR between CY25-30E
HBM Sales Forecast by Vendor ($bn)



Source: BofA Global Research estimates

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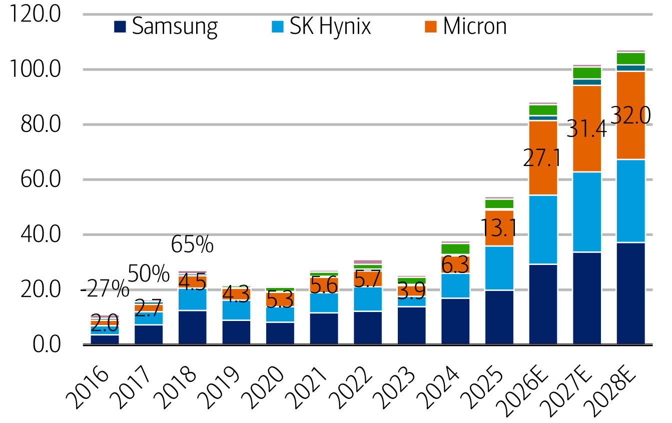
Structurally lower supply elasticity

Meanwhile, we also flag supply elasticity of memory is structurally increasingly lower due to limited capital, packaging, power, and geopolitics.

Despite increasing capex outlook for key memory vendors, we believe capacity increase is still generally limited through CY27-28 as capex is allocated towards building clean room space. For instance, MU guided its FY26 capex to be >$25bn vs. FY25 $13.8bn, with the majority of the increase for cleanroom facility. For FY27, the company expects construction related capex would be up $10bn YoY (rest of growth equipment spend).

We believe construction spend to generally outgrow equipment spend throughout CY26-27 (due to cleanroom space building), and meaningful capacity increase is likely in CY28 and beyond. For MU, its Idaho fab first output is expected to begin in mid-CY27 (CY28 ramp), while its Singapore advanced packaging facility for HBM is on track to contribute begin CY27 (CY28 full ramp).

Exhibit 24: DRAM Industry Capex Estimate ($bn)
We expect overall DRAM capex to meaningfully increase again in CY26-28



Source: BofA Global Research estimates

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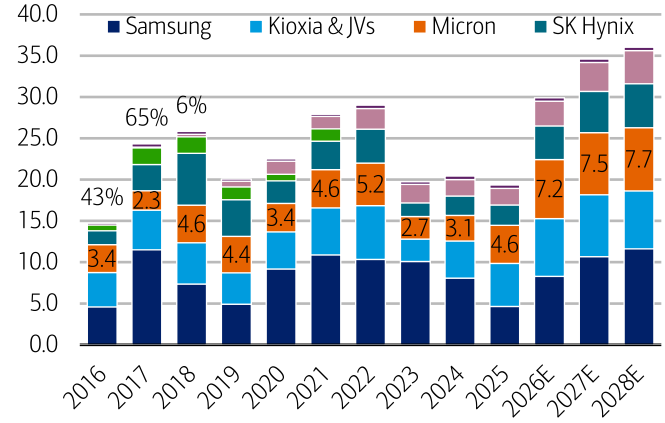
Exhibit 25: NAND Industry Capex Estimate ($bn)
We expect overall NAND capex to meaningfully increase again in CY26-28



Source: BofA Global Research estimates

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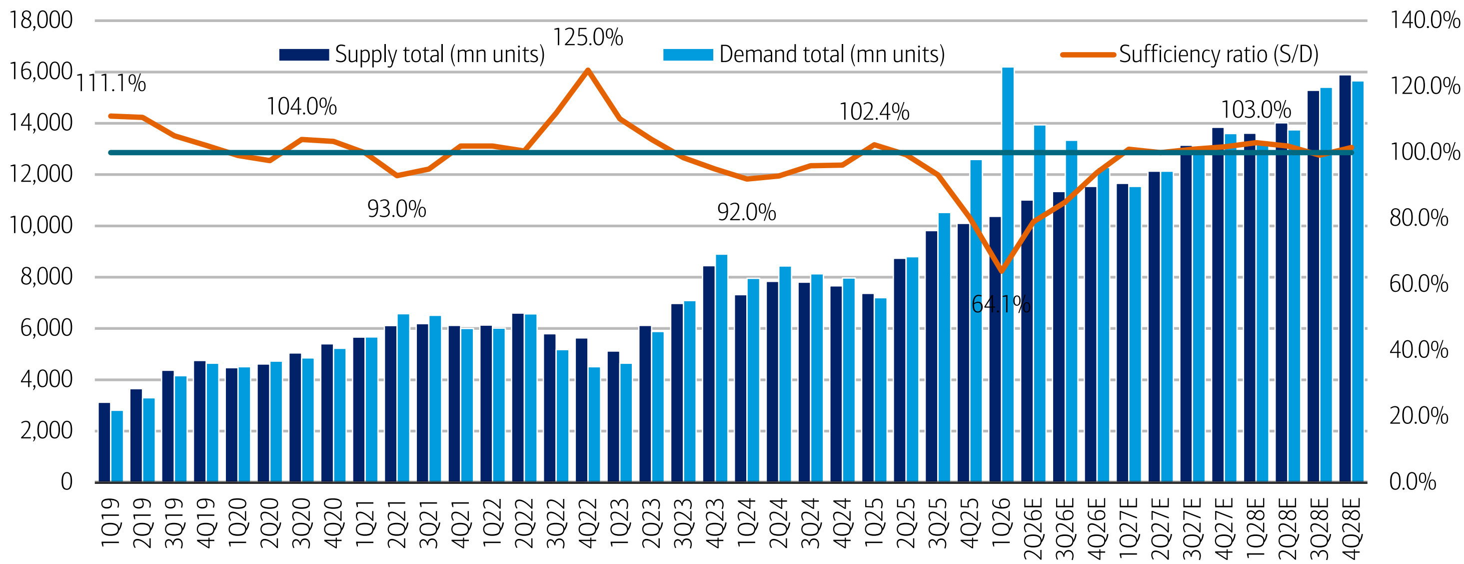
DRAM Supply/Demand Sustainability Ratio

Given DRAM industry supply and demand outlook, we see no material over-supply of DRAM (sufficiency ratio consistently >110%) through CY28, and we expect ASPs to remain relatively less cyclical vs. in prior down cycles.

Exhibit 26: We see no material over-supply of DRAM (sufficiency ratio consistently >110%) through CY28, expect ASPs to remain relatively less cyclical vs. in prior down cycles

DRAM Industry Supply/Demand and Sufficiency Ratio %



Source: BofA Global Research estimates

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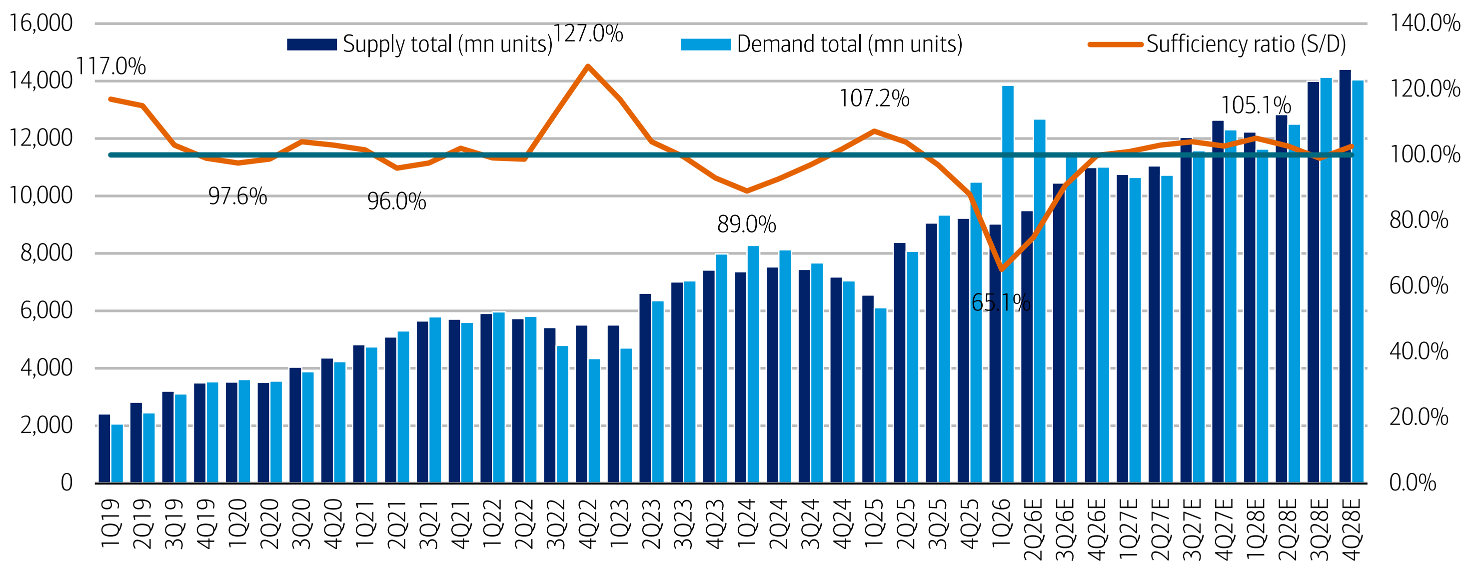
NAND Supply/Demand Sustainability Ratio

Similarly given NAND industry supply and demand outlook, we see no material over-supply of NAND (sufficiency ratio consistently >110%) through CY28, and we expect ASPs to remain relatively less cyclical vs. in prior down cycles.

Exhibit 27: We see no material over-supply of NAND (sufficiency ratio consistently >110%) through CY28, expect ASPs to remain relatively less cyclical vs. in prior down cycles

NAND Industry Supply/Demand and Sufficiency Ratio %



Source: BofA Global Research estimates

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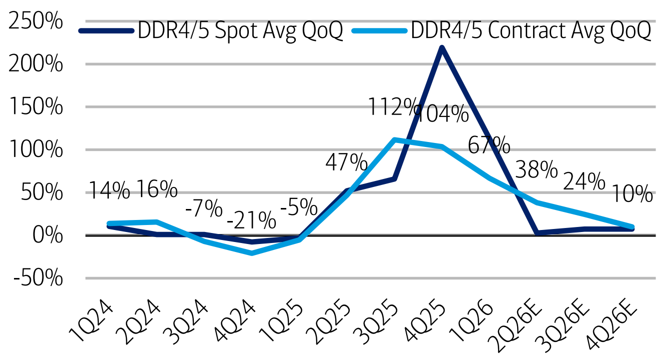
DRAM/NAND Pricing Trends

Following exceptionally strong spot/contract pricing trends over the last few months, we expect the growth to generally stabilize throughout the rest of 2026. However, we don't anticipate a QoQ decline until perhaps in 2027 given the current favorable supply/demand dynamics for both DRAM/NAND.

Exhibit 28: Strong DRAM spot/contract pricing trends throughout 2026
DRAM Spot/Contract ASP Trends



Source: BofA Global Research estimates

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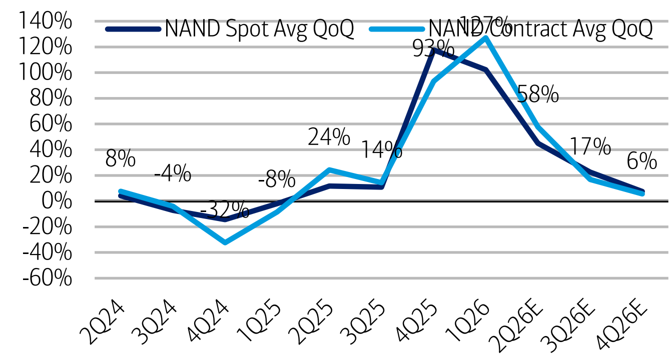
Exhibit 29: Strong NAND spot/contract pricing trends throughout 2026
NAND Spot/Contract ASP Trends



Source: BofA Global Research estimates

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MU Estimate Changes

Based on the latest pricing trends/outlook and MU's increased long-term sales/contract visibility, we raise our MU sales/EPS ests. Reiterate Buy on MU.

Exhibit 30: We raise MU FY26/27/28E sales by +11%/+27%/+31%, EPS by +21%/+35%/+43% to $97.78/$139.45/$139.29
MU Estimate Changes

	Sales ($mn)			Non-GAAP EPS		
	Old	**NEW**	**delta**	**Old**	**NEW**	**delta**
Nov-25	$13,643	$13,643	$0	$4.78	$4.78	$0.00
Feb-26	$23,860	$23,860	$0	$12.20	$12.20	$0.00
May-26E	$35,185	$38,429	$3,245	$20.44	$22.84	$2.40
Aug-26E	$39,754	$48,317	$8,563	$23.57	$29.88	$6.32
FY26E	**$112,441**	**$124,249**	**$11,808**	**$61.01**	**$69.73**	**$8.72**
YoY%	200.8%	232.4%	10.5%	636.0%	741.2%	14.3%
Nov-26E	$42,048	$52,689	$10,640	$24.94	$32.82	$7.89
Feb-27E	$42,786	$54,312	$11,526	$25.20	$33.77	$8.57
May-27E	$43,366	$56,324	$12,957	$25.32	$34.98	$9.66
Aug-27E	$45,495	$57,419	$11,923	$26.45	$35.40	$8.95
FY27E	**$173,696**	**$220,743**	**$47,047**	**$101.92**	**$136.99**	**$35.07**
YoY%	54.5%	77.7%	27.1%	67.1%	96.5%	34.4%
Nov-27E	$45,372	$57,803	$12,431	$25.97	$35.28	$9.32
Feb-28E	$41,893	$54,813	$12,920	$23.28	$32.91	$9.63
May-28E	$41,502	$54,998	$13,496	$22.77	$32.81	$10.04
Aug-28E	$44,493	$59,195	$14,701	$24.68	$35.59	$10.91
FY28E	**$173,261**	**$226,809**	**$53,548**	**$96.69**	**$136.60**	**$39.91**
YoY%	-0.3%	2.7%	30.9%	-5.1%	-0.3%	41.3%
CY26E	**$140,847**	**$163,295**	**$22,448**	**$81.17**	**$97.78**	**$16.61**
YoY%	232.9%	285.9%	15.9%	619.6%	766.9%	20.5%
CY27E	**$177,019**	**$225,857**	**$48,838**	**$102.95**	**$139.45**	**$36.50**
YoY%	25.7%	38.3%	27.6%	26.8%	42.6%	35.4%
CY28E	**$175,459**	**$232,052**	**$56,593**	**$97.16**	**$139.29**	**$42.13**
YoY%	-0.9%	2.7%	32.3%	-5.6%	-0.1%	43.4%

Source: BofA Global Research estimates

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MU Valuation Analysis, $1,500 PO

Exhibit 31: We see ~$1,047 value from MU's traditional DRAM/NAND business (2.5x P/B) and ~$465 from HBM business (31x PE), totaling ~$1,500/sh in total value
MU Sum-of-Parts Analysis

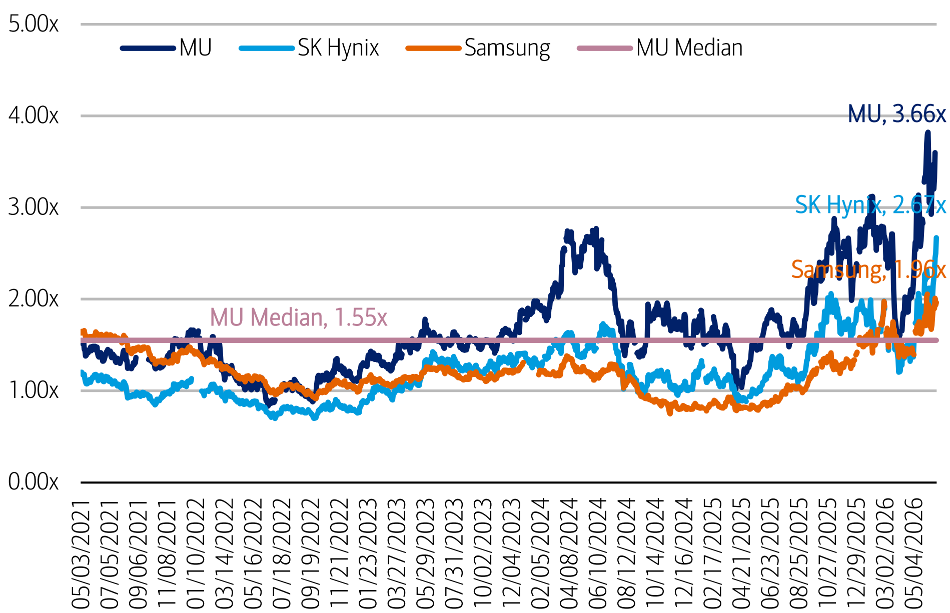
	CY25	CY26E	CY27E	CY28E	Notes
Trad. DRAM/NAND Business					
Total Book Value ($bn)	**58.8**	**171.2**	**332.6**	**494.8**	
HBM ($bn)	7.4	24.0	48.7	74.2	
HBM % (est.)	12.5%	14.0%	14.7%	15.0%	
Non-HBM ($bn)	51.5	147.2	283.9	420.6	
P/B Multiple				**2.5x**	Historical Upcycle Level
Value/sh				**$1,047**	
HBM Business					
Sales ($bn)	**6.9**	**13.2**	**21.3**	**30.0**	
OpM (est.)	50.0%	57.0%	67.0%	69.0%	
EBIT ($bn)	3.5	7.5	14.3	20.7	
Tax rate	13.2%	15.1%	15.1%	15.1%	
Net Income ($bn)	3.0	6.4	12.1	17.6	
DSO (mn)	1,145	1,150	1,157	1,163	
EPS ($)	$2.61	$5.55	$10.47	$15.10	
P/E Multiple				**31x**	AI Compute Peers Level
Value/sh				**$465**	
Total Sum-of-Parts				**$1,512**	

Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

MU is currently trading at 3.7x consensus P/B, above the upper-end of its historical range and its historical median of 1.6x (2yr fwd). However, we believe MU is now a fundamentally different company with a new and sustainable HBM-related business, in addition to its traditional commodity memory (DRAM/NAND) business.

Exhibit 32: MU is currently trading at ~3.66x fwd P/B vs. historical ~1.55x median. SK Hynix at ~2.67x and Samsung Electronics at ~1.96x
Major Memory Vendors 2-Yr Fwd P/B Valuation



Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

INTC: Raise PO to $160, Buy

We reiterate Buy on INTC, raise PO to $160 from $135. We highlight INTC's expanded opportunity in server CPUs which it can manufacture itself (gives higher supply visibility), as well as increased opportunities in its external foundry business.

Fully established IDM by CY30, EPS power $6+

We value INTC as a fully integrated IDM company with EPS power of ~$6.24 by CY30, with our new PO based on 31x PE (in-line with AI compute peers, vs. 25x prior given recent re-rating in peer multiples), discounted back two years.

We have moved from using CY28 sum-of-parts valuation methodology in the past which we now believe under-represents many of the company's CPU and foundry potentials that are further out in CY29-30+.

- **Products:** We now expect INTC server CPUs sales to reach ~$40bn+ by CY30E (vast majority of DCAI segment is server CPUs), or about ~25% share of $170bn TAM.

- **Foundry:** While we are not changing our base case foundry segment model until there is further clarity on specific deal details, we present the multiple deals INTC is currently engaged with, size the potential opportunities below. Opportunities span across: Apple M-Series wafers, MediaTek TPU wafers, Terafab IP/packaging, and potentially other ARM-based server CPUs, edge AI expansion into Client – all made more possible via recent CDNS 14A collab announcement.

Exhibit 33: We expect $6+ INTC EPS power by CY30, now raise PO to $160, versus $135 prior
INTC Sales Power, EPS Power, Valuation Methodology

	CY26E	CY27E	CY28E	CY29E	CY30E	CY30E Methodology		CAGR
Products	**$55.0**	**$61.4**	**$68.4**	**$77.6**	**$86.1**			**11.9%**
CCG	$31.6	$32.8	$34.4	$38.6	$42.4			
DCAI	$23.4	$28.6	$33.9	$39.0	$43.7	~25%	Share of $170bn TAM	
Foundry	**$1.1**	**$8.0**	**$17.5**	**$30.8**	**$47.1**			**158.4%**
A/P	$0.4	$1.9	$3.5	$4.8	$6.0	30.0%	Share of $20bn 2.5D/3D TAM	
Wafer	$0.5	$5.7	$11.4	$17.0	$23.1	~6-7%	Share of $380bn Wafer Foundry TAM	
Apple	$0.0	$2.5	$5.4	$8.0	$10.6	40.0%	Share of M-Series, 25% A-Series	
TPU/MediaTek	$0.5	$3.3	$6.0	$9.0	$12.4	100.0%	Share of TPU (MediaTek)	
Terafab	$0.1	$0.3	$1.5	$4.0	$8.0		IP, Packaging, Wafer (partial): mid-2028 begin	
Other HPC Products	$0.0	$0.0	$1.0	$5.0	$10.0		ARM-based server CPU opp'ty	
Other	**$2.3**	**$2.5**	**$2.5**	**$2.6**	**$2.8**			
Net Sales	**$58.4**	**$71.9**	**$88.3**	**$111.0**	**$135.9**			**23.5%**
Products EBIT	**$18.0**	**$21.6**	**$25.7**	**$30.8**	**$33.2**	**38.5%**	**vs. OpM Target (38.5%)**	
Foundry EBIT	**-$9.3**	**-$6.6**	**-$1.8**	**$3.1**	**$8.5**	**18.0%**	**vs. OpM Target (27.5%)**	
Other, Unallocated	-$1.1	-$1.0	-$1.0	-$1.0	-$1.0			
Total EBIT (non-GAAP)	**$7.6**	**$13.9**	**$22.8**	**$32.9**	**$40.6**			**52.0%**
OpM (%)	**13.0%**	**19.4%**	**25.8%**	**29.6%**	**29.9%**			
Interest Income	-$0.2	-$0.8	-$0.8	-$0.8	-$0.8			
EBT	$7.4	$13.2	$22.0	$32.1	$39.8			
Tax Rate	11.5%	12.0%	12.0%	12.0%	12.0%			
Net Income	$6.5	$11.6	$19.4	$28.3	$35.1			
NCI Adj.	-$1.1	-$1.2	-$1.2	-$1.2	-$1.3			
Net Income (non-GAAP)	**$5.4**	**$10.4**	**$18.2**	**$27.0**	**$33.8**			
EPS Power	**$1.06**	**$2.01**	**$3.47**	**$5.07**	**$6.24**			**55.9%**
Shares Outst.	5,106	5,172	5,248	5,326	5,406			
Multiple (P/E)					**31x**		AI Peer Average Level	
Implied valuation/Price Objective (CY28E)			**$160**	**$177**	**$195**			
Discount (WACC)			-9.4%	-9.4%				

Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

ARM: Raise PO to $460, Neutral

We see ARM as one of the most prominent beneficiaries of the rising server CPU tide, as the fastest overall ecosystem share gainer through CY30.

We continue to see the server CPU market as ~$170bn TAM by CY30E, see ARM's collective server CPU value share to reach ~50% by CY30E across: ~35% in merchant CPUs (NVDA, ARM, QCOM, etc.) and ~15% in custom CPUs (hyperscale ASIC such as AWS Graviton, GOOGL Axion, MSFT Cobalt).

However, at ~$420, we view the stock as generally fairly valued, with our new sum-of-parts valuation methodology yielding a PO of $460. Reiterate Neutral.

Sum-of-Parts Valuation: IP & Chip Businesses

We flag the immense opportunities ahead in standalone chip business (vs. historical only IP business), and value the company on a sum-of-parts basis.

- **IP business (~$342):** For the historical IP business, we value it based on ~2.5x PEG framework, in-line with high-growth semis and IP peers of 2x-3x.

  This is based on +25% Royalty sales CAGR (vs. +20% prior as we expect stronger ARM-based server CPU ramps such as QCOM's upcoming product) and +12% Licensing sales CAGR, which at 65% combined OpM would yield EPS power of $6+ by FY31E, or +28% EPS CAGR.

- **Chip business (~$118):** For the new chiplet business, we value based on ~31x PE, in-line with high-growth AI compute vendor average of ~31x.

  We see potential for ~$20bn chip sales power by FY31E (~CY30E), based on 12% share outlook of our ~$170bn CPU TAM forecast. We use 30% OpM target that ARM outlined (in-line with ~30% OpM for more established vendors such as AMD).

- **Total company:** For the total company valuation, we discount back two years and combine – **1) $342 worth of IP business**, and **2) $118 worth of chip business** – totaling **~$460 of total company value**.

Exhibit 34: We raise our PO to $460, versus $335 prior, based on CY30E sum-of-parts for IP and Chip businesses, discounted back 2 years
ARM Sales Power, EPS Power, Valuation Methodology

	FY26	FY31E - IP Only			FY31E - Chip Only			FY31E - Total Company		
IP Sales	**$4,920**	**$12,041**						**$12,041**		
Royalty Sales	$2,613	$7,975	25%	CAGR						
Licensing Sales	$2,307	$4,066	12%	CAGR						
Chip Sales	**$0**				**$20,400**			**$20,400**		
CPU TAM					$170,000	12%	TAM & Share			
Total Sales Power	**$4,920**	**$12,041**	20%	CAGR	**$20,400**	33%	CAGR	**$32,441**	46%	CAGR
IP EBIT	$2,115	$7,827	65%	OpM				$7,827	65%	OpM
Chip EBIT	$0				$6,120	30%	OpM	$6,120	30%	OpM
Total EBIT	**$2,115**	**$7,827**			**$6,120**			**$13,947**		
Interest Income	$107	$50			$50			$100		
EBT	$2,222	$7,877			$6,170			$14,047		
Tax Rate	15.0%	15.0%			15.0%			15.0%		
Net Income	**$1,889**	**$6,695**			**$5,245**			**$11,940**		
EPS Power	**$1.77**	**$6.09**	**28%**	**CAGR**	**$4.77**	**22%**	**CAGR**	**$10.85**	**44%**	**CAGR**
Shares Outst.	1,068	1,100			1,100			1,100		
~CY30 Valuation		**$427**	**2.5x**	**PEG**	**$148**	**31x**	**PE**	**$574**	**53x**	**Implied PE**
~CY29 Valuation		$382			$132			$514	-10.5%	Discount
~CY28 Valuation		**$342**			**$118**			**$460**	-10.5%	Discount

Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

CRDO: Raise PO to $340, Buy

We reiterate Buy on CRDO, raise PO to $340 from $252.

We highlight CRDO's continued ramp of its key AEC product line (100G/lane today, 200G/lane beginning CY28-29) at multiple hyperscalers (and emerging hyperscalers), as well as its expanding opportunity in optical DSPs, ZF optics, active LED cables (ALCs), and PCIe retimers – all of which are expected to begin ramping in the CY27-28 timeframe.

Given our recent discussion with the company (at 2026 BofA Global Tech Conference) and checks – which focused on CRDO's rock solid AEC market outlook over the next 5 years as well as CRDO's bulletproof reliability and incremental nature of its optical portfolio – we overall raise CRDO sales outlook by 2-11% for FY27-28E, EPS outlook by 5-15%.

$340 PO on 34x CY28E PE, <1x PEG

Our new $340 PO is now based on 34x CY28E PE vs. 33x CY27E prior, as we move to CY28E valuation to reflect CRDO's new product ramps expected in the out years..

This 34x multiple is still within the range of similar high-growth compute/optical semis peers trading between 22x-60x, and in-line with CRDO's historical 2-yr fwd trading range of 17x-244x. It also remains attractive, in our view, given just <1x PEG using our +42% EPS CAGR outlook (CY26-28E).

Exhibit 35: We raise CRDO FY27/28E sales by +2%/+11%, EPS by +5%/+15% to $5.93/$8.82
CRDO Estimate Changes

	Sales ($mn)			Non-GAAP EPS (ex. SBC)		
	Old	**NEW**	**delta**	**Old**	**NEW**	**delta**
1Q26	$223	$223	$0	$0.52	$0.52	$0.00
2Q26	$268	$268	$0	$0.67	$0.67	$0.00
3Q26	$407	$407	$0	$1.07	$1.07	$0.00
4Q26	$437	$437	$0	$1.16	$1.16	$0.00
FY26	**$1,335**	**$1,335**	**$0**	**$3.44**	**$3.44**	**$0.00**
YoY%	205.7%	205.7%	0.0%	385.7%	385.7%	0.0%
1Q27E	$470	$470	$0	$1.07	$1.07	$0.00
2Q27E	$506	$510	$4	$1.08	$1.10	$0.02
3Q27E	$668	$663	($5)	$1.59	$1.63	$0.04
4Q27E	$768	$815	$48	$1.90	$2.13	$0.22
FY27E	**$2,411**	**$2,458**	**$47**	**$5.65**	**$5.93**	**$0.28**
YoY%	80.6%	84.1%	2.0%	64.5%	72.7%	5.0%
1Q28E	$787	$852	$65	$1.90	$2.15	$0.25
2Q28E	$803	$886	$84	$1.90	$2.19	$0.29
3Q28E	$819	$922	$103	$1.90	$2.23	$0.33
4Q28E	$843	$958	$115	$1.95	$2.25	$0.30
FY28E	**$3,251**	**$3,618**	**$367**	**$7.65**	**$8.82**	**$1.17**
YoY%	34.9%	47.2%	11.3%	35.4%	48.6%	15.3%
CY26E	**$2,080**	**$2,080**	**($0)**	**$4.90**	**$4.96**	**$0.06**
YoY%	94.8%	94.7%	0.0%	86.5%	88.7%	1.2%
CY27E	**$3,176**	**$3,475**	**$299**	**$7.60**	**$8.69**	**$1.09**
YoY%	52.7%	67.1%	9.4%	55.2%	75.3%	14.3%
CY28E	**$3,716**	**$4,277**	**$561**	**$8.72**	**$9.98**	**$1.27**
YoY%	17.0%	23.1%	15.1%	14.6%	14.8%	14.5%

Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

MRVL: Raise PO to $365, Buy

We reiterate Buy on MRVL, raise PO to $365 from $240.

We highlight MRVL's continued ramp of its key connectivity product lines (optical DSPs 100G-200G/lane today, TIAs, drivers) on strong AI infra deployment outlook, coupled with upcoming adoption of Celestial AI-based CPO product lines.

Moreover, we hosted MRVL CEO Matt Murphy earlier this month at our 2026 BofA Global Tech Conference who exhibited further confidence in the ramps of two upcoming custom XPU projects (AWS and MSFT), alongside custom XPU attach projects that span NICs, CXL products, etc.

EPS Power $15+ by CY30E, see $365 PO

Overall, we continue to expect ~$15+ EPS power for MRVL by CY30, from current year ~CY26 EPS outlook of ~$4.

Our new $365 PO is based on 31x PE off our conceptual EPS power outlook of ~$15, in-line with AI compute peer forward average of 31x, discounted back two years at MRVL's estimated WACC of ~11%.

Exhibit 36: We expect $15+ MRVL EPS power by CY30, now raise PO to $365, versus $240 prior
MRVL Sales Power, EPS Power, Valuation Methodology

	CY25	CY26E	CY27E	CY28E	CY29E	CY30E	CAGR
Marvell EPS Power	**FY26**	**FY27E**	**FY28E**	**FY29E**	**FY30E**	**FY31E**	**'26-30E**
Total Sales ($mn)	**$8,195**	**$11,132**	**$16,525**	**$25,302**	**$31,253**	**$37,352**	**35%**
YoY (%)	*42.1%*	*35.8%*	*48.4%*	*53.1%*	*23.5%*	*19.5%*	
Data Center	**$6,100**	**$8,829**	**$14,153**	**$22,870**	**$28,700**	**$34,671**	**41%**
Connectivity	**$3,031**	**$4,740**	**$7,612**	**$10,092**	**$13,133**	**$16,149**	**36%**
Celestial AI	$0	$0	$305	$1,527	$3,027	$4,527	
Other Connectivity	$3,031	$4,740	$7,307	$8,564	$10,106	$11,622	25%
Custom-silicon	**$1,520**	**$2,007**	**$4,056**	**$10,068**	**$12,585**	**$15,302**	**66%**
XPU	$1,320	$1,557	$2,856	$6,868	$8,585	$10,302	60%
XPU attach	$200	$450	$1,200	$3,200	$4,000	$5,000	83%
Others (Storage, NICs, Switching, AEC, GP server)	**$1,550**	**$2,082**	**$2,485**	**$2,710**	**$2,981**	**$3,220**	12%
	$2,094	**$2,303**	**$2,372**	**$2,432**	**$2,553**	**$2,681**	**4%**
Gross Profit	$4,872	$6,520	$9,496	$14,220	$17,440	$20,694	33%
Gross Margin (%)	**59.5%**	**58.6%**	**57.5%**	**56.2%**	**55.8%**	**55.4%**	
Operating Income	$2,891	$4,151	$6,605	$10,978	$13,685	$16,505	41%
Operating Margin %	**35.3%**	**37.3%**	**40.0%**	**43.4%**	**43.8%**	**44.2%**	
pf-EPS Power ($)	**$2.85**	**$3.92**	**$6.11**	**$10.02**	**$12.49**	**$15.04**	**40%**
Shares (mn)	866	910	921	930	933	936	1%
Multiple (P/E)						**31x**	
Implied valuation/Price Objective (CY28)				**$365**	**$409**	**$459**	
Discount (WACC)				-10.8%	-10.8%		

Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

ALAB: Raise PO to $450, Neutral

Given our recent discussion with the company (at 2026 BofA Global Tech Conference) and checks, we raise ALAB sales outlook by 10-17% for FY27-28E, EPS outlook by 25-31%. Particularly, we flag material earnings leverage for ALAB in the near-term given the company's increasing scale in its new product lines (Scorpio Series and other custom fabrics such as for NVLink Fusion).

Exhibit 37: We raise ALAB FY26/27/28E sales by +2%/+10%/+17%, EPS by +6%/+25%/+31% to $3.10/$4.41/$5.78

ALAB Estimate Changes

	Sales ($mn)			Non-GAAP EPS		
	Old	NEW	delta	Old	NEW	delta
Mar-26	$308	$308	$0	$0.61	$0.61	$0.00
Jun-26E	$360	$360	$0	$0.69	$0.69	$0.00
Sep-26E	$399	$406	$8	$0.77	$0.81	$0.04
Dec-26E	$444	$467	$23	$0.86	$0.98	$0.12
FY26E	**$1,511**	**$1,542**	**$31**	**$2.94**	**$3.10**	**$0.16**
YoY%	77.2%	80.9%	2.0%	59.5%	68.2%	5.5%
Mar-27E	$463	$501	$38	$0.82	$1.00	$0.18
Jun-27E	$488	$533	$45	$0.85	$1.05	$0.21
Sep-27E	$526	$579	$53	$0.91	$1.14	$0.23
Dec-27E	$544	$612	$69	$0.93	$1.20	$0.27
FY27E	**$2,020**	**$2,225**	**$205**	**$3.51**	**$4.41**	**$0.89**
YoY%	33.7%	44.3%	10.1%	19.5%	42.2%	25.4%
Mar-28E	$567	$661	$95	$0.97	$1.30	$0.33
Jun-28E	$602	$715	$112	$1.07	$1.43	$0.35
Sep-28E	$640	$749	$109	$1.15	$1.49	$0.34
Dec-28E	$665	$780	$116	$1.21	$1.56	$0.35
FY28E	**$2,474**	**$2,906**	**$432**	**$4.41**	**$5.78**	**$1.37**
YoY%	22.5%	30.6%	17.4%	25.5%	31.2%	31.2%

Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

EPS Power $9+ by CY30E, see $450 PO

Specifically, we now see ALAB's new products (Scorpio and Custom) to make up ~$1.1bn in CY27E, ~$1.65bn in CY28E, ~$2.3bn in CY29E, and ~$3bn by CY30E, making up ~65%+ of the total company sales by the end of the decade.

By CY30E, we see total company **EPS power of $9+**, growing at ~31% CAGR.

For valuation, we raise our **PO to $450** from $240 prior, **now on 77x CY28E PE** vs. 66x CY27E prior, as we look out into the key Scorpio and new product ramps that are scheduled further out in the out years. This is also generally **in-line with our ~2.0x PEG framework** for CY28E, in-line with high growth AI compute/networking peers, given CY26-28E EPS CAGR outlook of +37%.

Our $450 PO (based on ~2.0x PEG framework on CY28E) is also in-line with 2.0x our CY30 EPS power of $9+ growing at ~31% CAGR, discounted back two years at ALAB's estimated WACC of ~11%.

See Exhibit 38 for details on ALAB's sales/EPS power forecasts.

Exhibit 38: We expect $9+ ALAB EPS power by CY30, now raise PO to $450, versus $240 prior
ALAB Sales Power, EPS Power, Valuation Methodology

	2026E	2027E	2028E	2029E	2030E	CAGR '26-'30
Aries	$709	$747	$785	$824	$866	5.1%
Taurus	$278	$361	$441	$552	$662	24.3%
Leo	$13	$21	$33	$49	$66	50.5%
Scorpio	$542	$1,036	$1,337	$1,762	$2,185	41.7%
P-Series	*$413*	*$603*	*$679*	*$829*	*$953*	*23.2%*
X-Series / UALink	*$129*	*$433*	*$657*	*$934*	*$1,232*	*75.9%*
Other/Custom	$0	$60	$310	$560	$810	
Total Sales ($mn)	**$1,542**	**$2,225**	**$2,906**	**$3,747**	**$4,589**	**31.3%**
Gross Profit	$1,129	$1,594	$2,070	$2,662	$3,260	
GM %	**73.2%**	**71.6%**	**71.2%**	**71.0%**	**71.0%**	
Opex	$528	$685	$824	$995	$1,150	
Opex %	34.3%	30.8%	28.4%	26.6%	25.1%	
EBIT	$601	$909	$1,246	$1,667	$2,111	
OpM %	**39.0%**	**40.8%**	**42.9%**	**44.5%**	**46.0%**	
Net Financial Income	$46	$46	$46	$46	$46	
Pre-tax income	$648	$955	$1,293	$1,714	$2,157	
Tax Rate	11.8%	12.0%	12.0%	12.0%	12.0%	
Non-GAAP Net Income	**$571**	**$840**	**$1,138**	**$1,508**	**$1,898**	
Non-GAAP EPS	**$3.10**	**$4.41**	**$5.78**	**$7.44**	**$9.09**	**30.9%**
YoY (%)	*68%*	*42%*	*31%*	*29%*	*22%*	
Shares Outst.	184.4	190.8	196.8	202.8	208.8	
Multiple (PEG)					**2.0x**	
Implied valuation/Price Objective (CY28E)			**$450**	**$503**	**$562**	
Discount (WACC)			-11%	-11%		

Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

TER: Raise PO to $525, Buy

We reiterate Buy on TER, raise our PO to $525 (from $365) on 41x CY28E PE (41x CY27E PE prior) as we roll the valuation year forward and raise FY27/28E sales estimates +2%/+7% and FY27/28E pf-EPS estimates by +9%/+16%, reflecting higher industry forecasts and improved confidence in TER's ability to gain meaningful share at NVDA vs. incumbent sole-source Advantest.

At our Global Tech Conference in June, TER framed the 2025 ATE TAM as roughly $9bn, while discussing how a $250bn WFE environment could "easily" support a $20bn ATE TAM. A $20bn 2028 ATE TAM would imply ATE as a % of total WFE increasing slightly from 7.7% in 2025 to 8.1% (based on our $245bn CY28E forecast), which we see as justified by rising test intensity and complexity but slightly offset by lower testing memory torque vs. logic.

With TER's evergreen model of $6bn revenue within a $12-$14bn ATE TAM (which would imply ATE intensity vs. WFE declining to 4.9%-5.7%), our raised $6.6bn CY28E revenue estimate may prove conservative. However, with TER expecting 2H

Further, all l-t drivers remain intact with 1) TER able to achieve ~30% 2nd source share at NVDA (~$50m exp. in 2H within multi $bn NVDA ATE TAM) if they are able to execute on product qualification and performance, 2) TER expected to hold ~50% share of custom ASIC test despite QoQ lumpiness, 3) rising HBM/DRAM test intensity, 4) CPO/SiPho driving further TAM expansion with $100m in 2026 rising to $300-$700m by 2028, and 5) strong robotics growth driven by primary customer Amazon expanding automation.

However, TER's presence on the back-end results in visibility remaining low vs. front-end WFE peers while 2H is expected to be down -20% HoH due to custom ASIC tester shipment timing variability and so while we remain optimistic on the potential, we keep estimates relatively in check until we get an update on 2H26 and 2027 visibility from TER's 2Q earnings in July.

Exhibit 39: We raise FY27E/28E sales estimates +2%/+7% to $5.5bn/$6.6bn and raise FY27/28E pf-EPS +9%/+16% to $9.75/$12.84

Summary of estimate changes reflecting higher WFE/industry forecasts and improved confidence in TER's ability to gain share vs. Advantest at NVDA

Model Changes

	Sales ($mn)			EPS (Non GAAP, incl SBC)		
	Old	**NEW**	**delta**	**Old**	**NEW**	**delta**
1Q26E	$1,282	$1,282	$0	$2.56	$2.56	$0.00
2Q26E	$1,201	$1,201	$0	$2.01	$2.01	$0.00
3Q26E	$897	$990	$94	$0.82	$1.30	$0.49
4Q26E	$937	$1,000	$63	$0.92	$1.34	$0.41
FY26E	**$4,317**	**$4,473**	**$156**	**$6.32**	**$7.22**	**$0.90**
YoY%	35.3%	40.2%	3.6%	59.7%	82.4%	14.2%
1Q27E	$1,120	$1,158	$38	$1.59	$1.82	$0.23
2Q27E	$1,261	$1,281	$20	$2.01	$2.18	$0.18
3Q27E	$1,376	$1,395	$19	$2.33	$2.50	$0.17
4Q27E	$1,603	$1,630	$27	$3.00	$3.24	$0.24
FY27E	**$5,361**	**$5,465**	**$104**	**$8.92**	**$9.75**	**$0.82**
YoY%	24.2%	22.2%	1.9%	41.3%	35.1%	9.2%
1Q28E	$1,583	$1,689	$106	$2.93	$3.36	$0.43
2Q28E	$1,501	$1,612	$111	$2.67	$3.11	$0.44
3Q28E	$1,492	$1,601	$109	$2.62	$3.05	$0.43
4Q28E	$1,578	$1,694	$116	$2.86	$3.32	$0.46
FY28E	**$6,154**	**$6,595**	**$441**	**$11.08**	**$12.84**	**$1.77**
YoY%	14.8%	20.7%	7.2%	24.1%	31.8%	16.0%

Source: BofA Global Research estimates

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ACLS: Raise PO to $156, Underperform

VECO already generally priced in at EPS power $9+

The merger with VECO is on track for 2H26 close, subject to regulatory approval (China the only one left). Given recent VECO outlook increase (InP laser optical component wins), we now see potential pro-forma EPS power of $9+ by CY28 (see Exhibit below), up from ~$5.50 for CY27E when the deal was first announced in Oct-2025.

While our standalone ACLS estimates have remained generally unchanged at ~$4.40 CY27E EPS, we acknowledge the potential accretion from the merger, raise PO to $156 from $130, now on 26x CY28E PE (vs. 27x CY27E), near the bottom of WFE peers trading between 24x-43x, or conceptually using unchanged 16x PE at ~$9 EPS power – in-line with ACLS historical 2-yr forward median of 16x. ACLS stock at the current >30x CY28 PE (or 19x using $9 EPS power) is already fully valued, in our view. Reiterate Underperform.

ACLS/VECO pf-EPS power $9+ by CY28

On Oct 1, ACLS and VECO announced intent to merge to create the fourth largest US-based semicap equipment vendor. We view the proposed deal as overall positive for ACLS, with potential product/revenue synergies across 1) front-end – ACLS ion implant and VECO annealing; 2) wide bandgap materials – ACLS SiC and VECO MOCVD (GaN-on-SiC); 3) foundry/logic – ACLS trailing-edge and VECO leading-edge for new customer acquisitions; and 5) back-end – advance packaging.

For instance, by offering ACLS's ion implant and VECO's annealing together, ACLS aims to deliver a more complete/integrated solution and ultimately penetrate VECO's leading-edge customers such as TSMC – where ACLS currently has no exposure. Specifically, ACLS views the potential penetration/addressable opportunity of ion implant into advanced logic to be ~$400mn TAM, for which they currently represent <5%.

Mgmt. believes the proposed deal to be accretive to non-GAAP EPS within the first 12 months of close, and we now see potential EPS power up to ~$9 by CY28.

Exhibit 40: We expect the proposed deal to be ~28.3%/~59.3%/~63.3% accretive to non-GAAP pro-forma potential EPS in CY26/27/28E

ACLS/VECO Potential EPS Accretion vs. Dilution Math

	FY23	FY24	FY25	Hypothetical ACLS FY26E				Hypothetical ACLS FY27E				Hypothetical ACLS FY28E			
	ACLS	ACLS	ACLS	ACLS	VECO	Adj.	pf	ACLS	VECO	Adj.	pf	ACLS	VECO	Adj.	pf
Sales ($mn)	$1,131	$1,018	$839	$846	$444	$13	$1,303	$925	$1,076	$39	$2,041	$1,018	$1,292	$65	$2,375
COGS	$638	$561	$460	$487	$246	$6	$738	$525	$591	$17	$1,133	$571	$698	$26	$1,294
Gross profit	$493	$457	$379	$359	$199	$7	$565	$401	$485	$22	$907	$448	$594	$39	$1,080
Gross margin (%)	43.6%	44.9%	45.2%	42.4%	44.7%	56.0%	43.3%	43.3%	45.0%	56.8%	44.5%	44.0%	46.0%	60.2%	45.5%
Opex	$207	$220	$221	$237	$113	($5)	$345	$260	$256	($16)	$500	$270	$310	($25)	$555
Operating income	$285	$237	$159	$122	$85		$220	$141	$229		$407	$178	$283		$525
Operating margin (%)	25.2%	23.3%	18.9%	14.4%	19.2%		16.9%	15.2%	21.2%		20.0%	17.5%	21.9%		22.1%
Interest expense	($13)	($19)	($19)	($14)	($3)	$0	($17)	($16)	($8)	$0	($24)	($16)	($10)	$0	($26)
Pre-tax income	$298	$257	$178	$136			$237	$157			$431	$194			$550
Tax rate (%)	11.8%	12.8%	13.2%	14.8%			14.8%	15.0%			15.0%	15.0%			15.0%
Tax expense	$35	$33	$24	$20			$35	$24			$65	$29			$83
Net income	$263	$224	$154	$116			$202	$133			$366	$165			$468
Sharecount (mn)	33	33	32	31		11	42	30		22	52	30		22	52
EPS ($)	$7.93	$6.84	$4.86	$3.77			$4.84	$4.40			$7.01	$5.53			$9.03
Accretion/dilution ($)							$1.07				$2.61				$3.50
Accretion/dilution (%)							28.3%				59.3%				63.3%

Source: BofA Global Research estimates, Visible Alpha

BofA GLOBAL RESEARCH

Semicap PO changes and LT EPS power

Advanced Energy (AEIS)

We raise AEIS to $450 from $430, now based on 32x CY28E PE vs. 36x CY27E prior to reflect further product/project opportunities in semicap/data center that are expected to ramp beginning CY27-28. Our 32x multiple still within historical 11x-36x range, justified by AEIS' rising exposure to fast-growing semicap and data center power markets, with improving margin and cash flow profiles. Reiterate Buy.

Applied Materials (AMAT)

We raise AMAT to $720 PO from $540, now based on 36x CY28E PE vs. 32x CY27E prior to reflect strong potential earnings power of ~$27 by CY28E (see Exhibit 40), 35% above current consensus modeling. Our new PO credits the potential for broad-based WFE share gains, especially in DRAM and potential for gross margin expansion into the mid-50% justifying a basis near the high-end of AMAT's 10x-42x historical forward PE range. Reiterate Buy.

Lam Research (LRCX)

We raise LRCX to $480 PO from $330, now based on 47x CY28E PE vs 36x CY27E prior to reflect strong potential earnings power of ~$15 by CY28E, 42% above current consensus modeling. Our higher PO factors in ongoing share gains across foundry/logic, DRAM, and advanced packaging, while also reflecting unique leverage to NAND upgrades and greenfield capacity expansion. We also think a multiple re-rating is warranted given a step up in gross margins towards the mid-50% range hence justifying valuation closer to the high-end of a historical 9x-50x forward PE range. Reiterate Buy.

KLA Corporation (KLAC)

We raise KLAC to $317 PO from $210 now based on 53x CY28E PE vs. 40x CY27E prior as the mix of WFE spending could shift favorably for KLAC into CY27/28E given higher foundry/logic, more design starts, and process control intensive tech inflections. We also expect strong gains in advanced packaging inside a $10-$15bn SAM LT which will help KLAC take greater share of overall WFE. At $250bn WFE, EPS could be +50% higher than what consensus is currently modeling per our EPS power analysis below which justifies a multiple at the high-end of a 12-53x historical range. Reiterate Buy.

MKS Instruments (MKSI)

We raise MKSI to $500 from $380, now on 22x CY28E EV/EBITDA vs. 18x CY27E prior. This new PO and multiple reflects the strong earnings leverage ahead as Semi Market should benefit from surging WFE investment through CY28 where MKSI's products are especially levered to etch/dep inflections while Electronics and Packaging is helped by robust AI substrate/PCB demand where rising complexity is the main driver. Free cash flow margins should expand which can help pay down debt to lower net leverage. Reiterate Buy.

Exhibit 41: We see potential EPS upside of 35%/42%/51% for AMAT/LRCX/KLAC if the industry reaches $250bn WFE by CY28E

EPS power of AMAT, LRCX, KLAC at $250bn WFE

Units in $bn	AMAT	LRCX	KLAC
CY28 WFE		$250	
Share	20.0%	16.0%	9.7%
Systems Revenue	$50.0	$40.0	$24.3
Services	$7.9	$9.5	$4.3
Other	$1.7	-	-
Total	**$57.9**	**$49.5**	**$28.5**
Gross Profit	$30.1	$26.5	$18.3
Gross Margin	52.0%	53.5%	64.0%
Opex	$7.5	$5.9	$4.3
Opex % of sales	13%	12%	15%
Operating Income	$22.6	$20.5	$14.0
Operating Margin	39.0%	41.5%	49.0%
Interest Income/Expense	$0.0	($0.0)	($0.1)
Pre-Tax Income	$22.6	$20.5	$13.9
Tax Rate	11.0%	14.0%	14.5%
Net Income	$20.1	$17.6	$11.9
pf-EPS	**$26.62**	**$15.29**	**$9.06**
Share Count	756	1154	1309
Consensus CY28 EPS	**$19.80**	**$10.77**	**$6.00**
Potential EPS upside (%)	34.5%	41.9%	51.0%
Forward PE Assumption	27.0x	32.0x	35.0x
Implied CY28E Val'n	$719	$489	$317
CY25-28E Sales CAGR	27.1%	34.0%	11.5%
CY25-28E EPS CAGR	41.4%	46.2%	22.8%

Source: BofA Global Research estimates, company reports, Bloomberg, Visible Alpha

BofA GLOBAL RESEARCH

Glossary:

2.5D: Two-and-a-Half-Dimensional

3D: Three-Dimensional

4F²: Four-F-Squared

14A: Intel 14A Process Node

18A: Intel 18A Process Node

A/P: Advanced Packaging

ACLS: Axcelis Technologies

ADI: Analog Devices

ADR: American Depositary Receipt

AEC: Active Electrical Cable

AEIS: Advanced Energy Industries

AGI: Artificial General Intelligence

AI: Artificial Intelligence

ALAB: Astera Labs

ALC: Active LED Cable

ALGM: Allegro MicroSystems

AMAT: Applied Materials

AMBA: Ambarella

AMBQ: Ambiq Micro

AMD: Advanced Micro Devices

ARM: Arm Holdings

ASIC: Application-Specific Integrated Circuit

ASP: Average Selling Price

AWS: Amazon Web Services

AVGO: Broadcom

EDA: Electronic Design Automation

EUV: Extreme Ultraviolet

EV: Electric Vehicle / Enterprise Value

F/L: Foundry/Logic

FinFET: Fin Field-Effect Transistor

FLOPs: Floating-Point Operations per Second

GaN: Gallium Nitride

GB: Gigabyte

GFS: GlobalFoundries

GM: Gross Margin

GOOGL: Alphabet

GPU: Graphics Processing Unit

HBM: High-Bandwidth Memory

HBM4: Fourth-Generation High-Bandwidth Memory

HBM5: Fifth-Generation High-Bandwidth Memory

High-K: High-Dielectric-Constant

High-NA: High Numerical Aperture

HPC: High-Performance Computing

IDM: Integrated Device Manufacturer

IMF: International Monetary Fund

INTC: Intel

InP: Indium Phosphide

IP: Intellectual Property

KLAC: KLA Corporation

KV Cache: Key-Value Cache

LITE: Lumentum Holdings

LLC: Limited Liability Company

LPO: Linear Pluggable Optics

LPDDR: Low-Power Double Data Rate

LPU: Language Processing Unit

LRCX: Lam Research

LRO: Linear Receive Optics

LSCC: Lattice Semiconductor

LT: Long-Term

LTA: Long-Term Agreement

LTAs: Long-Term Agreements

LV: Light Vehicle

MCHP: Microchip Technology

MCU: Microcontroller Unit

MKSI: MKS Instruments

MOCVD: Metal-Organic Chemical Vapor Deposition

MPU: Microprocessor Unit

MRVL: Marvell Technology

MSFT: Microsoft

MTSI: MACOM Technology Solutions

MU: Micron Technology

NA: Numerical Aperture / Not Available

NAND: NOT-AND Flash Memory

NIC: Network Interface Card

nm: Nanometer

NVDA: NVIDIA

NVFP4: NVIDIA 4-Bit Floating Point

NVLink: NVIDIA High-Speed Interconnect

NVMI: Nova

NXPI: NXP Semiconductors

ON: onsemi

PC: Personal Computer

PCIe: Peripheral Component Interconnect Express

QLC: Quad-Level Cell

QCOM: Qualcomm

RISC-V: Reduced Instruction Set Computer Five

SBC: Stock-Based Compensation

SiC: Silicon Carbide

SIA: Semiconductor Industry Association

SK: SK Hynix

SNPS: Synopsys

SoC: System-on-Chip

SOTP: Sum-of-the-Parts

SRAM: Static Random Access Memory

SSD: Solid-State Drive

SWKS: Skyworks Solutions

TAM: Total Addressable Market

TB: Terabyte

TER: Teradyne

TIA: Transimpedance Amplifier

TPU: Tensor Processing Unit

TSMC: Taiwan Semiconductor Manufacturing Company

TSV: Through-Silicon Via

TXN: Texas Instruments

UALink: Ultra Accelerator Link

VECO: Veeco Instruments

WFE: Wafer Fab Equipment

WOLF: Wolfspeed

wspm: Wafer Starts Per Month

x86: Intel/AMD Processor Architecture

XPU: X Processing Unit

ZF: Zero-Flap

CAMT: Camtek

CDNS: Cadence Design Systems

CEO: Chief Executive Officer

CCG: Client Computing Group

CMP: Chemical Mechanical Planarization

COHR: Coherent Corp

CPO: Co-Packaged Optics

CPU: Central Processing Unit

CRDO: Credo Technology

CXL: Compute Express Link

CY: Calendar Year

DCAI: Data Center and AI

DC: Data Center

DRAM: Dynamic Random Access Memory

DSP: Digital Signal Processor

Exhibit 42: Stocks mentioned
Rating and price summary

BofA Ticker	Bloomberg ticker	Company name	Price	Rating
AEIS	AEIS US	Advanced Energy	US$ 388.23	B-1-7
AMAT	AMAT US	Applied Materials	US$ 640.18	B-1-7
ARM	ARM US	Arm Holdings	US$ 407.72	C-2-9
ALAB	ALAB US	Astera Labs	US$ 439.66	C-2-9
ACLS	ACLS US	Axcelis Technologies	US$ 183.92	C-3-9
CRDO	CRDO US	Credo Technology	US$ 302.52	C-1-9
INTC	INTC US	Intel	US$ 140.94	C-1-9
KLAC	KLAC US	KLA Corp	US$ 269.16	B-1-7
LRCX	LRCX US	Lam Research	US$ 409.54	C-1-7
MRVL	MRVL US	Marvell	US$ 307.86	C-1-7
MU	MU US	Micron	US$ 1211.38	C-1-7
MKSI	MKSI US	MKS Instruments	US$ 420.56	C-1-7
TER	TER US	Teradyne	US$ 457	C-1-7

Source: BofA Global Research

BofA GLOBAL RESEARCH

Price objective basis & risk

Advanced Energy Industries (AEIS)

Our $450 PO is based on 32x CY28E non-GAAP EPS. Our PO basis is above the historical median of 22x and toward the upper end of historical 11x-36x, justified by AEIS's rising exposure to fast-growing semicap and data center power markets, with improving margin and cash flow profiles.

Upside risks: 1) faster than anticipated industrials market share gains, 2) faster than anticipated gross margin expansion from cyclical recovery.

Downside risks: 1) increasing exposure to more commoditized (less proprietary) end markets, 2) potential broadening of China export restrictions impacting semicap equipment customers, 3) historically cyclical nature of semiconductor capital spending.

Applied Materials, Inc. (AMAT)

Our PO of $720 is based on 36x our CY28E P/E estimate. Our PO basis is near the high-end of AMAT's historical 10-42x trading range justified given the potential for WFE outgrowth in 2026/27E and remains at a discount to other large peers given more balanced growth profile and lower profitability.

Downside risks to our PO are: ongoing US government probe that we are unable to size the financial impact at this time, slower-than-expected capital spending cycle, delay in memory capacity adds, market share loss in deposition, implant, thermal, CMP, etch, or process control segments, merger & integrations risk, and macro headwinds.

Arm Holdings (ARM)

Our $460 PO is based on conceptual sum-of-parts valuation that values the IP business at $342 (2.5x CY30E PEG, discounted back 2 years) and the chiplet business at $118 (31x CY30E PE, discounted back 2 years), with both multiples in-line with competitive peer range/average. Our PO implies 130x total company CY28E PE which is still within ARM's historical 35x-147x range, given longer-term data center content and silicon/chiplet (AGI CPU) opportunities, offset by high SoftBank dependence and opex ramp.

Upside risks: 1) better than expected smartphone/PC/server unit shipment, 2) faster adoption of higher-royalty rate v9 and CSS IP at customers, 3) faster share gains and ramp of AGI CPU chiplet lineup, 4) further proliferation of AI data centers and hyperscale-specific custom products, 5) Qualcomm/Nuvia content expansion postsettlement, 6) small floating volume.

Downside risks: 1) historically cyclical nature of semiconductor units, 2) high exposure to mature smartphone market, 3) competition against established x86 and other custom Arm-based CPU offerings in the data center, 4) emerging competition from RISC-V in low-end consumer markets, 5) rising geopolitical tensions and deterioration of Arm China relationship, 6) ongoing Qualcomm/Nuvia litigation, 7) small trading float.

Astera Labs Inc (ALAB)
Our $450 PO is based on 77x CY27E P/E. Our PO basis is above the high-end of similar high-growth compute/optical semiconductor peers trading at 18x-68x but in-line with historical 2.0x PEG framework given ALAB's superior sales growth outlook.

Downside risks to our PO are: (1) project delays/changes at leading customer Amazon, (2) increased competition from large cap connectivity peers Marvell/Broadcom, (3) delayed adoption of UALink-based scale-up switches or active electrical cable (AEC) products, (4) downturn in spending across hyperscalers and network operators, (5) Inability for ALAB to scale and meet demand from products beginning to ramp, (6) supply chain headwinds limiting available capacity.

Upside risks are: (1) unforeseen, accelerated engagements in PCIe- or UALink-based switching products, (2) a faster than anticipated ramp of lead customer accelerator projects leading to higher than expected ALAB products demand, (3) GM consistency/expansion over time, overcoming headwinds from an ongoing mix shift to more hardware/module products vs. chips/silicon

Axcelis Technologies (ACLS)
Our PO of $156 is based on 26x our CY28E non-GAAP EPS, adjusted for net cash. This is toward the low-end of peers trading at 25x-43x, but well above ACLS historical median of 17x. We view this is justified given potential synergies and increased revenue potential from the proposed Veeco merger, offset by concerns around near-term auto/EV demand/utilization and ACLS's outsized China exposure.

Downside risks: 1) financial/integration/regulatory risks associated with the closure of proposed merger with Veeco, 2) potential expansion of restrictions of tool shipments from US to China which would impact its trailing-edge portfolio, 3) strong competition from larger vendors such as AMAT, 4) historically cyclical nature of semiconductor capital spending

Upside risks: 1) faster than anticipated recovery in China SiC market demand, with a faster depletion of used tools inventory, 2) restructuring actions at certain US-based SiC suppliers potentially revitalizing competition in the market and creating more customers for ACLS, 3) greater sales synergy outlook in key ion implant + annealing, and wide bandgap (SiC + GaN) markets from the potential VECO merger, and 4) ongoing shortage in conventional DRAM (non-HBM) creating sharp demand for new ion implant tools

Credo Technology (CRDO)
Our $340 PO is based on 34x CY28E P/E. Our PO basis is within the range of similar high-growth compute/optical semiconductor peers trading 18x-66x.

Downside risks to our PO are: (1) increased competition from large cap peers Marvell/ Broadcom, (2) delayed adoption of active electrical cable (AEC) products, (3) downturn in spending across hyperscalers and network operators, (4) Inability for CRDO to scale and meet demand from products beginning to ramp, (5) supply chain headwinds limiting available capacity.

Upside risks are: (1) unforeseen, accelerated AEC engagements, (2) sudden rebound in ramp for higher margin optical DSP products, (3) prudent opex/margin management to support EPS in downcycle.

Intel (INTC)

Our $160 price objective is based on 31x CY30E EPS power of $6+, discounted back two years, given the company's key server CPU and external foundry wafer/packaging opportunities that extend far out. Our PO implies 10.7x our total company 2028E EV/S which is well above historical 1.7x-4x range but we view as appropriate given increasing server CPU and external foundry opportunity, offset by near-/medium-term manufacturing ramp-up uncertainties.

Upside risks to our price objective are 1) key external foundry packaging/wafer deals that could significantly boost sales/utilization, 2) greater than expected yields/ramps at 18A and upcoming 14A nodes, resulting in greater GM/utilization profile, 3) stronger than expected PC market from Windows 10 refresh or AI uplift, 4) geopolitical tensions boosting sentiment for domestic manufacturing asset.

Downside risks to our price objective are 1) lower than yield/ramp at Intel Foundry, particularly for its new 18A and upcoming 14A nodes, 2) lack of material external foundry customer in wafer processing, 3) weaker-than-expected trends in a mature PC market, which is largest revenue generator for Intel, 4) accelerated share loss to major CPU competitors.

KLA Corporation (KLAC)

We assign a $317 PO based on 53x CY27E P/E. Our PO multiple is at the high end of its 12x-53x historical range justified by KLAC's leading profit margin, longer lead times (resulting in greater visibility), and less cyclical topline supports a slightly high multiple vs. semicap peers.

Downside risks to our PO are the cyclical nature of the semiconductor capital spending and its impact on earnings, competitive price and market share issues, ability to get new products and technologies into the market in a timely manner.

Lam Research Corp. (LRCX)

We assign a $480 PO based on 47x CY27E PE. Our PO basis is in the upper half of the historical 9x-50x trading range justified by ongoing memory and leading-edge foundry/logic WFE cycle, high-teens EPS CAGR over time, etch/deposition product leadership, rising etch/deposition intensity, share gains, growing foundry/logic exposure over memory, improving prospects of NAND recovery, and robust FCF generation, offset by near-term concerns around cost inflation and tariffs.

Upside risks are tech inflections, F/L share gains, NAND upgrades.

Downside risks are slower than expected capital spending cycle, delay in memory capacity adds, market share loss in etch or clean segments, merger & integrations risk, macro headwinds, customer consolidation and China.

Marvell Technology, Inc. (MRVL)

Our $365 price objective is based on 31x CY30E EPS power of $15+ (including stock-based comp), discounted back two years. Our PO basis is modestly higher than 26x historical median but within 14x-47x range and is justified by an improvement in visibility for major customer ASIC projects, broadening AI portfolio across connectivity/switch/compute, and cyclical industry risks offset by ASIC upside, networking strength, and AEC/CPO/scale-up share gains.

Upside risks: 1) Faster than anticipated ramp/visibility in major custom ASIC projects, 2) Continued growth in DSP-based pluggable market, versus new LPO/LRO techs, 3) Share gains in emerging AEC/CPO/scale-up switch markets against incumbents.

Downside risks: 1) Loss of visibility in key custom ASIC projects, particularly in the next-gen 3nm/2nm chips at AWS and Microsoft, 2) Competition in AI compute, with merchant vendors continuing to proliferate and ASIC incumbent AVGO winning many of the large hyperscaler/AI customers, 3) cyclical industry risks including potential slowdown in legacy storage, enterprise networking, carrier markets.

Micron Technology, Inc (MU)
Our $1500 PO is based on a sum-of-parts valuation that values: (1) traditional cyclical memory business at $1,040/sh at 2.5x CY28E P/B, toward the high-end of MU's long-term range 0.8x-3.1x as we are potentially in a memory upcycle, and (2) AI HBM business at 31x CY28E PE, in-line with AI compute peer group median.

Downside risks: (1) larger than expected memory ASP decline, (2) greater competition from China newcomers, (3) share loss to large competitors, (4) softening of demand across major end markets such as data center, smartphones, or PCs.

MKS Instruments (MKSI)
Our PO of $500 PO is based on 22x our CY28E EV/EBITDA. Our PO multiple is near peers trading at 23x at the median and justified given MKSI's higher margins offset by higher leverage and higher non-Semi exposure.

Downside risks: 1) historically cyclical nature of semiconductor capital spending, 2) potential broadening of China export restrictions impacting semicap equipment customers, 3) above-average debt profile following Atotech acquisition.

Teradyne (TER)
Our $525 price objective is based on 41x PE applied to our CY28E Non-GAAP EPS. Our PO basis is towards the upper end of TER's 13x-47x historical trading range and justified given TER's elevated growth, share gain potential, and long-term robotics optionality.

Upside risks to our price objective are (1) accelerated adoption of leading edge compute/networking/memory test equipment, (2) improved positioning in the compute market, (3) strong momentum at Apple in the semitest business, (4) faster than expected ramp of Amazon Robotics arm.

Downside risks to our PO are weaker and lumpy AI revenue materialization from ASIC/memory customers, extended auto/industrial weakness, mobile/2nm adoption unfavorable mix, merchant GPU testing win delay/not materializing.

Analyst Certification

We, Vivek Arya, Duksan Jang and Michael Mani, hereby certify that the views each of us has expressed in this research report accurately reflect each of our respective personal views about the subject securities and issuers. We also certify that no part of our respective compensation was, is, or will be, directly or indirectly, related to the specific recommendations or view expressed in this research report.

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BofA Securities is currently acting as joint financial advisor to ABB Ltd in connection with the sale of its Robotics Division to SoftBank Group Corp, which was announced on October 8, 2025.

US - Semiconductors and Semiconductor Capital Equipment Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	Advanced Energy Industries	AEIS	AEIS US	Duksan Jang
	Advanced Micro Devices, Inc	AMD	AMD US	Vivek Arya
	Allegro Microsystems	ALGM	ALGM US	Vivek Arya
	Analog Devices Inc.	ADI	ADI US	Vivek Arya
	Applied Materials, Inc.	AMAT	AMAT US	Vivek Arya
	Broadcom Inc	AVGO	AVGO US	Vivek Arya
	Cadence	CDNS	CDNS US	Vivek Arya
	Camtek	CAMT	CAMT US	Michael Mani
	Credo Technology	CRDO	CRDO US	Vivek Arya
	Intel	INTC	INTC US	Vivek Arya
	KLA Corporation	KLAC	KLAC US	Vivek Arya
	Lam Research Corp.	LRCX	LRCX US	Vivek Arya
	M/A-Com	MTSI	MTSI US	Vivek Arya
	Marvell Technology, Inc.	MRVL	MRVL US	Vivek Arya
	Microchip	MCHP	MCHP US	Vivek Arya
	Micron Technology, Inc	MU	MU US	Vivek Arya
	MKS Instruments	MKSI	MKSI US	Michael Mani
	Nova	NVMI	NVMI US	Michael Mani
	NVIDIA Corporation	NVDA	NVDA US	Vivek Arya
	onsemi	ON	ON US	Vivek Arya
	Synopsys	SNPS	SNPS US	Vivek Arya
	Teradyne	TER	TER US	Vivek Arya
	Texas Instruments Inc.	TXN	TXN US	Vivek Arya
NEUTRAL				
	Ambarella	AMBA	AMBA US	Vivek Arya
	Arm Holdings	ARM	ARM US	Vivek Arya
	Astera Labs Inc	ALAB	ALAB US	Vivek Arya
	Coherent Corp	COHR	COHR US	Vivek Arya
	Lumentum Holdings	LITE	LITE US	Vivek Arya
	NXP Semiconductors NV	NXPI	NXPI US	Vivek Arya
UNDERPERFORM				
	Axcelis Technologies	ACLS	ACLS US	Duksan Jang
	GlobalFoundries	GFS	GFS US	Vivek Arya
	Lattice Semiconductor	LSCC	LSCC US	Duksan Jang
	Qualcomm	QCOM	QCOM US	Vivek Arya
	Skyworks Solutions, Inc.	SWKS	SWKS US	Vivek Arya
RSTR				
	Ambiq Micro, Inc.	AMBQ	AMBQ US	Vivek Arya
RVW				
	Wolfspeed Inc	WOLF	WOLF US	Vivek Arya

Disclosures

Important Disclosures

Equity Investment Rating Distribution: Technology Group (as of 31 Mar 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	234	58.50%	Buy	123	52.56%
Hold	90	22.50%	Hold	43	47.78%
Sell	76	19.00%	Sell	23	30.26%

Equity Investment Rating Distribution: Global Group (as of 31 Mar 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	1993	55.76%	Buy	1186	59.51%
Hold	821	22.97%	Hold	509	62.00%
Sell	760	21.26%	Sell	400	52.63%

[R1] Issuers that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.

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Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster[R2]
Buy	≥ 10%	≤ 70%
Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

[R2]Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

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